A comparative machine learning framework for stellar subdwarf classification using spectral metrics and feature engineering

Ana V. Ojeda[a], Christopher A. Theissen[b], C. A. Guerrero[c], Juan J. Tapia[a], and T. Zhou[d]

[a]Instituto Politécnico Nacional, Centro de Investigación y Desarrollo de Tecnología Digital, Tijuana, B.C., 22435, México
[b]Department of Astronomy & Astrophysics, University of California, San Diego, La Jolla, CA 92093, USA
[c]Universidad Autónoma de Nuevo León, Facultad de Ciencias Físico Matemáticas, San Nicolás de los Garza, Nuevo León, 66451, México
[d]Department of Physics and Astronomy, Brigham Young University, Provo, UT 84602, USA

ABSTRACT

Automated classification of spectra from low-mass stars and brown dwarfs requires methods that remain reliable not only for spectral-type assignment but also for gravity- and metallicity-sensitive peculiar classes. This study revisits the automated classification of M0–T9 dwarfs observed with the SpeX prism spectrograph ($R \sim 120$, 0.8–2.5 μm), with emphasis on peculiar classes associated with surface gravity and metallicity. Rather than optimizing primarily for spectral type, the model-selection procedure prioritizes peculiar-class balanced accuracy (BAcc) to reduce the effect of the uneven representation of field, β, γ, d/sd, sd, and esd objects. A 90-dimensional input vector was constructed from binned spectral fluxes, targeted spectral slopes, the apparent J- and H-band magnitudes, and residual features measured relative to field standards of the same spectral type. Random forest (RF), support vector machine (SVM), K-nearest neighbors (KNN), and one-dimensional convolutional neural network (CNN) models were trained on 66,000 synthetic spectra, and their configurations were selected using a validation set of 396 spectra. Final performance was measured on an independent observed test set containing 1,300 spectra. KNN achieved the strongest overall performance, with 89.84% peculiar-class BAcc, 88.08% peculiar-class accuracy, and spectral-type accuracies of 96.85% and 99.54% within ±1 and ±3 spectral types, respectively. RF ranked second in peculiar-class BAcc, whereas CNN produced the lowest peculiar-class BAcc and SVM obtained the lowest spectral-type accuracy within ±3 spectral types. The RF feature-importance analysis identified the field-referenced features D_i, A_i, and K_i, the apparent J- and H-band magnitudes, several near-infrared spectral slopes, and spectral bins near 0.89–0.99 μm as the most informative variables for peculiar-class classification. These findings show that KNN provides the most effective class-balanced solution for the proposed representation, while RF offers complementary interpretability of the spectral and photometric information driving the classification.

Keywords: Machine learning, M/L dwarfs, Near-infrared astronomy, Subdwarfs, Low-metallicity stars.

1. INTRODUCTION

Stars exhibit diverse physical properties that can be investigated through spectroscopy. Spectral classification organizes stars according to similarities in their spectra and provides a consistent framework for comparing stellar populations.[1] Among low-mass stars, M dwarfs dominate the nearby stellar census.[2] Together with brown dwarfs, these objects are commonly referred to as cool dwarfs. The sample examined in this study comprises near-infrared (NIR) spectra of M, L, and T dwarfs spanning spectral types M0–T9.[3–5]

Further author information: (Send correspondence to A.A.A.)
A.A.A.: E-mail: aojeda@citedi.mx, Telephone: +52 664 506 0715

Spectral type alone, however, does not capture the full diversity of cool-dwarf spectra. Surface gravity modifies several NIR features, including iron hydride (FeH) and vanadium oxide (VO) absorption, alkali lines, and the shape of the H-band continuum.[6] Gravity-sensitive classification schemes use the suffixes β and γ to identify intermediate- and low-gravity objects, respectively.[7,8] Metallicity also affects spectral morphology. In M dwarfs, the relative strengths of titanium oxide (TiO) and calcium hydride (CaH) bands have been used to distinguish dwarfs, subdwarfs, extreme subdwarfs, and ultrasubdwarfs.[9,10]

Accordingly, the present study considers six gravity- and metallicity-dependent categories: field, β, γ, d/sd, sd, and esd. Here, field refers to objects classified as normal field dwarfs without an assigned low-gravity or metal-poor designation; d/sd denotes an intermediate dwarf/subdwarf class, sd denotes subdwarfs, and esd denotes extreme subdwarfs. These six categories are collectively referred to here as *peculiar classes*.

Traditional spectral classification commonly relies on visual comparison with spectra of established spectral standards,[3–5] as well as on diagnostic spectral indices and equivalent-width measurements.[5,6,11] These approaches form the basis of empirical classification systems, but their applicability and precision depend on spectral resolution, signal-to-noise ratio (SNR), wavelength coverage, and the availability of measurable diagnostic features.[6,11] Because spectral indices and equivalent widths characterize selected features or wavelength intervals, they do not necessarily incorporate the complete spectral morphology. Automated methods that analyze the full spectrum therefore provide a complementary approach for processing large spectroscopic samples consistently.[12–14]

Machine-learning (ML) methods have been applied to photometric surveys to estimate stellar parameters and extract astrophysical information from heterogeneous observational data.[15,16] Within cool-dwarf spectroscopy, ML methods have increasingly been used for automated identification, gravity-sensitive classification, and the estimation of spectral and physical properties from low-resolution spectra.[12,13,17] By analyzing patterns across multiple wavelength regions simultaneously, these methods provide a scalable complement to traditional feature-based classification approaches.

The present study builds directly on the work of Zhou et al., who applied Random Forest (RF), Support Vector Machine (SVM), and K-Nearest Neighbors (KNN) models to low-resolution ($\lambda/\Delta\lambda \approx 120$) NIR spectra of M0–T9 dwarfs obtained with the SpeX spectrograph at the Infrared Telescope Facility (IRTF).[14,18] Their baseline representation consisted of 67 average-flux features generated over 0.89–2.45 μm using 0.02 μm wavelength bins and excluding regions strongly affected by telluric absorption from Earth's atmosphere.[14] These features were used for spectral-type and peculiar-class classification with RF, SVM, and KNN models.

The primary objective of the present work is to determine whether peculiar-class classification can be improved through feature engineering, defined here as the construction of physically motivated descriptors from the spectra. Continuum slopes calculated across selected wavelength intervals are incorporated to represent broad spectral-shape variations associated with surface gravity and metallicity. Their effectiveness is evaluated using revised RF, SVM, and KNN configurations, while a convolutional neural network (CNN) is included as a complementary approach that learns relevant patterns directly from the binned spectral sequence. The selected models are evaluated using validation data and an independent test sample to assess their generalization beyond model development. Accuracy (Acc) and balanced accuracy (BAcc) are considered for peculiar-class and spectral-type classification, while spectral-type predictions are additionally evaluated within ±1 and ±3 spectral types. Their performance is compared with the corresponding configurations reported by Zhou et al., with particular emphasis on improving the identification of the β, γ, d/sd, sd, and esd categories.

2. METHODOLOGY

This study extends the machine-learning framework developed by Zhou et al.[14] to investigate whether the classification of peculiar classes can be improved through additional preprocessing, physically motivated feature engineering, and alternative supervised-learning configurations. Particular emphasis was placed on the discrimination of gravity- and metallicity-dependent categories that were less reliably classified in the reference study, including β, γ, d/sd, sd, and esd objects.

The original binned-flux representation used by Zhou et al.[14] was supplemented with continuum slopes calculated over selected wavelength intervals, apparent J- and H-band magnitudes (m_J and m_H), and residual

features measured relative to field standards of the same spectral type. These additional features were designed to represent spectral and photometric information potentially associated with surface gravity and metallicity. Missing photometric measurements were addressed through catalog retrieval and statistical imputation.

RF, SVM, KNN, and CNN classifiers were evaluated using the complete 90-dimensional input vector. Model selection prioritized peculiar-class BAcc on the validation set, and the selected configurations were subsequently evaluated using an independent observed test sample. Performance was measured using Acc and BAcc for peculiar-class and spectral-type classification, together with spectral-type BAcc within ±1 and ±3 spectral types.

2.1 The Data

To enable a direct comparison with the results reported by Zhou et al.,[14] this study used the same dataset and supporting files made publicly available in their GitHub repository.* For practical purposes throughout the remainder of this paper, the study by Zhou et al.[14] is referred to as the *reference work*.

2.1.1 Training set constructed from spectral standards

Zhou et al.[14] constructed the original training set from 70 low-resolution near-infrared spectral standards observed with the SpeX spectrograph mounted on the 3-m NASA Infrared Telescope Facility.[18] The prism-mode spectra have a resolving power of approximately $R \sim 120$ and cover the wavelength range from 0.8 to 2.5 μm. The data were accessed through the SpeX Prism Library Analysis Toolkit (SPLAT).[19]

The 70 spectral standards were selected from established classification sequences: normal field-dwarf standards defined by Kirkpatrick et al.,[20] surface-gravity standards defined by Allers and Liu, Gagné et al., and Cruz et al.,[6,8,21] and subdwarf standards defined by Burgasser and Greco et al.[22,23] The normal field standards span spectral types M0–T9, while the intermediate- and low-gravity standards classified as β and γ, respectively, are mainly available for late-M and L dwarfs. The metal-poor standards classified as d/sd, sd, and esd are primarily available for M and early-L types. Consequently, the training sample does not include T-dwarf standards belonging to the non-field peculiar classes.

The standard spectra have SNR values ranging from 2.5 to 975, with a mean of 141 and a median of 69. These values were calculated with SPLAT as the median SNR of the spectral pixels whose fluxes lie within the upper half of the flux distribution.[19] A complete list of the 70 standards, including their spectral types, object names, SNR values, and references, is provided in Table 1 of Ref. 14.

To generate a sufficiently large and balanced training set, Zhou et al.[14] produced 1,000 synthetic spectra from each standard by applying Gaussian perturbations to the measured spectral fluxes. This procedure resulted in an original training set of 70,000 synthetic spectra, with each generated spectrum retaining the spectral-type and peculiar-class labels of its corresponding parent standard.

2.1.2 Validation and test sets

According to Zhou et al.,[14] an independent sample of 1,548 near-infrared spectra was assembled for validation and testing. This sample consisted of 1,527 spectra from the SpeX Prism Library and 21 additional observations of metal-poor dwarfs not included in that library. Among the additional observations, 18 correspond to unique sources and three are repeated observations acquired at different epochs. The complete list of these additional metal-poor sources is provided in Table 3 of Ref. 14.

To illustrate the spectral diversity of the observed sample, Fig. 1 reproduces the spectral-standard visualizations reported by Zhou et al.[14] Panel (a) shows the field M-, L-, and T-dwarf spectral standards, panel (b) shows the low- and intermediate-gravity objects, and panel (c) shows the metal-poor objects.

From the complete sample, 248 observed spectra spanning the M, L, and T spectral classes were assigned to the validation set. An additional 148 synthetic spectra were included to improve the representation of spectral types with limited observational coverage, resulting in a validation set of 396 spectra. The remaining 1,300 observed spectra were reserved as the independent test set and were not used during model training or hyperparameter selection.[14]

*https://github.com/StellarDataLab/SpecAI

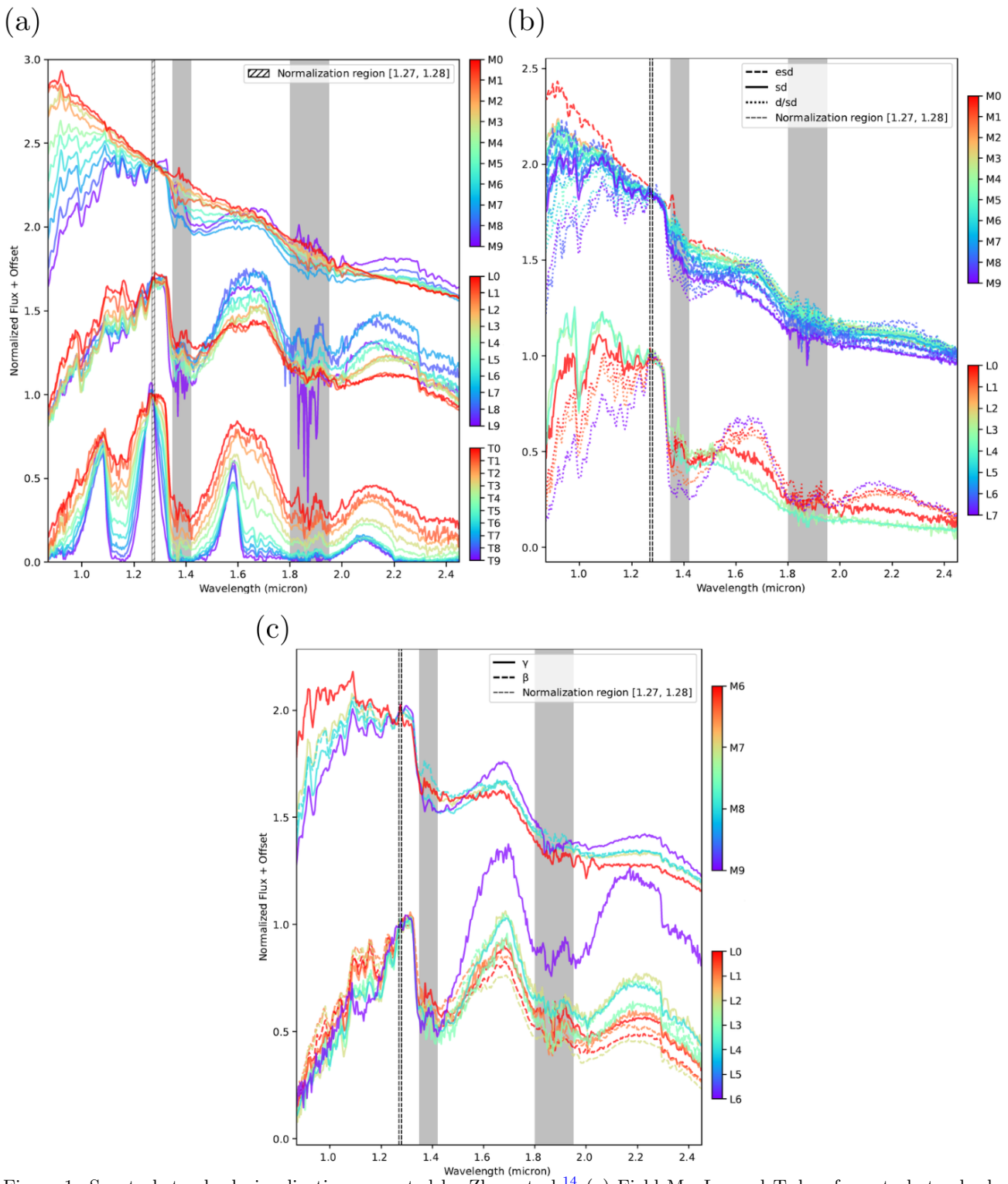


Figure 1: Spectral-standard visualizations reported by Zhou et al.[14] (a) Field M-, L-, and T-dwarf spectral standards. (b) Low- and intermediate-gravity objects. (c) Metal-poor objects. The spectra are normalized over the 1.27–1.28 μm region, and the shaded regions correspond to masked telluric absorption intervals. Reproduced from Zhou et al.[14]

The distribution of spectral types within these sets is not uniform. Early-M dwarfs are less represented because their spectral energy distributions peak toward optical wavelengths, where established classification schemes are commonly applied. Late-L dwarfs and objects near the L/T transition are also less abundant because of their intrinsic faintness and relative rarity. The combined spectral-type and peculiar-class distributions of the validation and test samples are shown in Fig. 1 of Ref. 14, while further details on the construction of these sets are provided in Secs. 2.2 and 3.1.5 of the same reference.

2.2 Dataset Assembly and Preprocessing

Before performing feature engineering and missing-data imputation, the original training, validation, and test sets were reorganized to establish a consistent structure across the three datasets. Dedicated metadata files were created to associate each spectrum with its object name, spectral-type label, peculiar-class label, apparent J- and H-band magnitudes, photometric uncertainties, and binned spectral fluxes.

The spectral preprocessing by Zhou et al. consisted of a sequence of quantitative steps applied consistently to all spectra. The retained wavelength coverage extended from 0.89 to 2.45 μm, encompassing the principal near-infrared photometric regions: the J-band at approximately 1.1–1.4 μm, the H-band at approximately 1.5–1.8 μm, and the K-band at approximately 2.0–2.4 μm. Each spectrum was normalized by the median flux measured within the 1.27–1.28 μm interval of the J-band.[14]

Additionaly, the wavelength intervals from 1.35 to 1.42 μm and from 1.80 to 1.95 μm were masked because they are strongly affected by telluric absorption, that is, absorption produced by molecules in Earth's atmosphere, primarily water vapor, which reduces atmospheric transmission and makes the measured flux less reliable. The remaining wavelength regions were divided into 0.02 μm bins, producing 67 average-flux features for each spectrum. The spectral-type labels were standardized through χ^2 comparison with the corresponding spectral standards. Further details regarding this preprocessing procedure are provided in Sec. 3 of Ref. 14.

2.2.1 Training set

Using the spectral standards, we created a dedicated spectral-standard metadata file containing the information required for the subsequent construction of the training set. For each standard, the apparent J- and H-band magnitudes and their corresponding photometric uncertainties were obtained through SPLAT,[19] together with the associated near-infrared spectrum. The spectra were processed following the procedure of Zhou et al.,[14] including normalization, masking of the telluric regions, and generation of the spectral flux bins. The resulting file contains the object name, spectral-type label, peculiar-class label, apparent J- and H-band magnitudes, photometric uncertainties, and binned spectral fluxes.

To maintain consistency with the labeling scheme adopted by Zhou et al.,[14] spectral types containing decimal values were rounded down to the nearest integer type. For example, an object classified as esdM7.5 was assigned the label esdM7. This reference file was subsequently used to associate the photometric magnitudes and their uncertainties with the corresponding parent standards in the synthetic training set.

The esdM6.5, sdM9.5, sdL3.5, and sdL4 spectral standards were removed from the reference file because these spectral types were not represented in either the validation or test set. Consequently, the 1,000 synthetic spectra generated from each of these standards were also removed from the training set, reducing its size to 66,000 synthetic spectra.

In addition, because the M6γ spectral standard TWA 8B was not included in the SPLAT catalog, its apparent J- and H-band magnitudes and corresponding uncertainties were retrieved from the SIMBAD astronomical database and incorporated into the spectral-standard metadata file.[24]

2.2.2 Validation and test sets

We created a dedicated metadata file for the validation and test samples using `SPL_data_7_15.csv`, obtained from the GitHub repository provided by Zhou et al.,[14] as the initial source. This file contains information for 1,527 spectra from the SpeX Prism Library (SPL), including object names, spectral-type labels, peculiar-class labels, signal-to-noise ratios (SNR), apparent J- and H-band magnitudes, and bibliographic references.

The 21 additional metal-poor observations used by Zhou et al.[14] were not included in the original file; therefore, their corresponding metadata were retrieved through SPLAT and appended to our file. The resulting metadata table contains the information required to associate the complete sample of 1,548 validation and test spectra with their corresponding objects and observational properties.

The repository also includes the file `SPL.csv`, which contains the spectral bins and associated labels for the complete sample of 1,548 spectra used to construct the validation and test sets. This file was essential because the validation and test files provided in the `data` directory contain only the binned spectra and their corresponding spectral-type and peculiar-class labels. They do not include the object names or explicitly identify which entries in the validation set correspond to observed or synthetic spectra. Therefore, `SPL.csv` was used to relate the spectra to their corresponding sources and to reconstruct the composition of the validation and test samples.

2.3 Spectral Feature Engineering

Starting from the preprocessed spectra described in Sec. 2.2, two complementary groups of spectral features were computed to emphasize differences associated with peculiar class. The first group consisted of targeted slope-based features designed to quantify broad changes in spectral shape across selected wavelength intervals. The second group consisted of field-reference residual features computed relative to the field spectral standard of the same spectral type, thereby quantifying the deviation of each spectrum from the corresponding field morphology.

2.3.1 Targeted Slope-Based Features

For this stage, the normalized spectral standards were organized into the M0–M9, L0–L9, and T0–T9 spectral-type sequences. A mean spectrum was calculated for each sequence to facilitate the visual identification of broad and local continuum variations. Figure 2 shows the T0–T9 sequence as a representative example of the visually guided slope-selection procedure. The corresponding M0–M9 and L0–L9 sequences are presented in Appendix A, Fig. 16(a) and Fig. 16(b), respectively.

Three broad analysis-defined intervals were selected: $\mathcal{R}_{\mathrm{YJ}} = [0.90, 1.34]$ μm, $\mathcal{R}_{\mathrm{H}} = [1.43, 1.79]$ μm, and $\mathcal{R}_{\mathrm{K}} = [1.96, 2.38]$ μm. These intervals are not formal photometric passbands; they cover portions of the spectrum commonly associated with the YJ, H, and K regions. Additional local slopes were selected to characterize shorter-scale spectral variations.

The intervals 0.90–1.08 μm and 1.08–1.12 μm overlap the region in which Zhou et al.[14] identified TiO and FeH features near 0.89 and 1.0 μm as having the highest feature-importance scores. These molecular features are sensitive to effective temperature, metallicity, and surface gravity.[21] Additional intervals were defined from 1.17 to 1.26 μm and from 1.26 to 1.34 μm.

In the H-band region, the intervals 1.43–1.585 μm, 1.43–1.69 μm, 1.585–1.63 μm, 1.585–1.69 μm, 1.585–1.79 μm, and 1.69–1.79 μm were considered. Several of these intervals overlap the 1.47–1.67 μm continuum region associated with surface-gravity variations.[14] Finally, the K-band region, intervals 1.96–2.08 μm, 2.08–2.21 μm, and 2.08–2.38 μm were included. These intervals cover the CH_4 and CO absorption regions near 2.20 and 2.30 μm, respectively, as well as a broader continuum affected by collision-induced H_2 absorption. Zhou et al.[14] found the K-band to be particularly relevant for surface-gravity classification, consistent with the gravity-sensitive behavior of this spectral region[6] and with previously defined H_2OD and H_2O-2 indices.[25,26]

For each wavelength pair, the spectral slope was calculated as

$$S_{\lambda_1,\lambda_2} = \frac{F(\lambda_2) - F(\lambda_1)}{\lambda_2 - \lambda_1}, \tag{1}$$

where $F(\lambda_1)$ and $F(\lambda_2)$ represent the normalized fluxes at the initial and final wavelengths, respectively.

The ability of the selected slopes to characterize differences among the M, L, and T spectral classes was evaluated using a one-way analysis-of-variance (ANOVA) F statistic.[27,28] For each slope, the statistic was calculated from the between-class and within-class variability as

$$F = \frac{\frac{1}{K-1}\sum_{g=1}^{K} n_g \left(\overline{S}_g - \overline{S}\right)^2}{\frac{1}{N-K}\sum_{g=1}^{K}\sum_{i=1}^{n_g} \left(S_{ig} - \overline{S}_g\right)^2}, \tag{2}$$

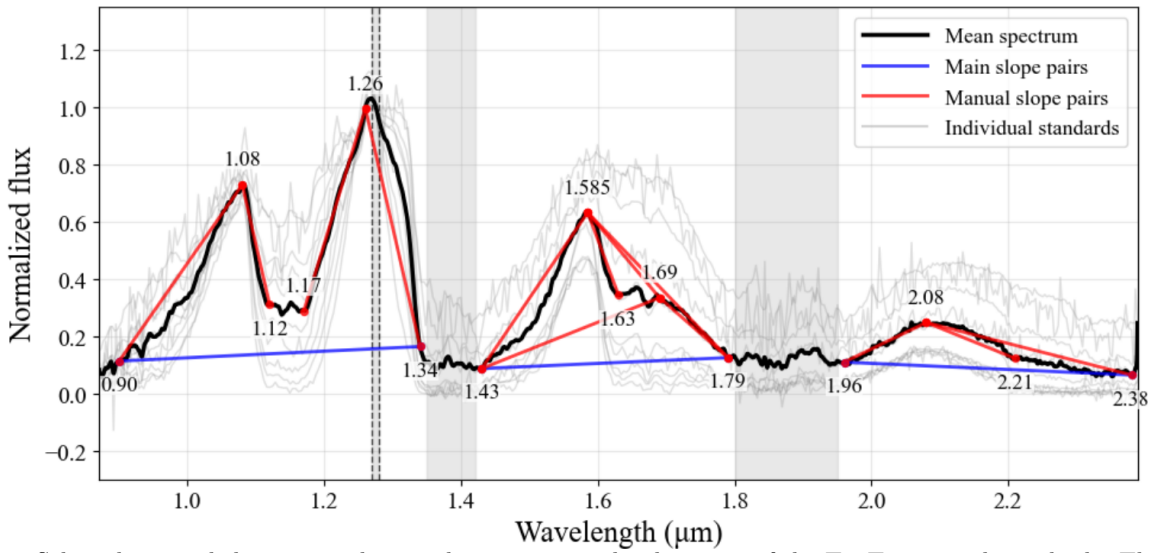


Figure 2: Selected spectral-slope intervals over the mean normalized spectra of the T0–T9 spectral standards. The three broad intervals are shown in blue, and the additional local intervals are shown in red. The gray shaded regions indicate the masked telluric absorption intervals, while the vertical dotted lines delimit the normalization interval.

where $K = 3$ corresponds to the M, L, and T spectral classes, S_{ig} is the slope of the ith spectral-type standard in class g, n_g is the number of standards in that class, $\overline{S}_g$ is its mean slope, and $\overline{S}$ is the overall mean slope. Here, $n_{\rm M} = n_{\rm L} = n_{\rm T} = 10$, corresponding to M0–M9, L0–L9, and T0–T9, respectively, and $N = 30$. Larger F values indicate greater between-class separation relative to the variability within each spectral class. The 16 targeted slopes were ranked in decreasing order of F, and the four highest-ranked intervals are shown in Fig. 3.

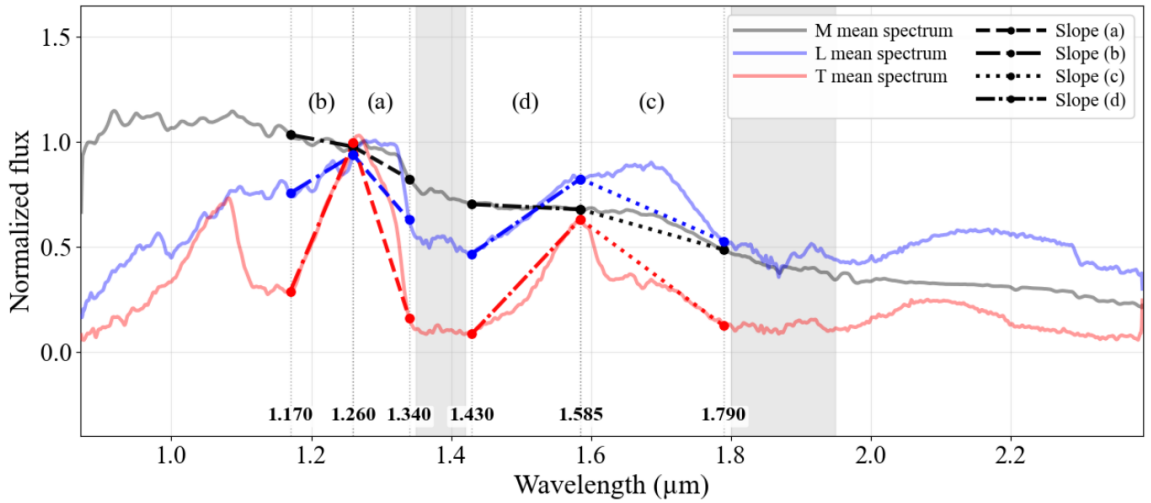


Figure 3: Four targeted slopes with the highest F statistics, shown over the mean M-, L-, and T-class spectra: (a) 1.260–1.340 μm, $F = 66.598$; (b) 1.170–1.260 μm, $F = 61.966$; (c) 1.585–1.790 μm, $F = 55.578$; and (d) 1.430–1.585 μm, $F = 54.654$. The shaded regions indicate the masked telluric absorption intervals.

2.3.2 Continuum-Removed Residual Features

To motivate the definition of the field-referenced residual features, we visually examined how spectral morphology changes when each spectrum is expressed relative to the field spectral standard of the same spectral type. This representation suppresses the dominant spectral-type dependence and emphasizes residual differences potentially associated with peculiar class.

Figure 4 illustrates this comparison for spectral type M4. The M4 d/sd, sd, and esd spectral standards, together with the M4 objects in the independent test set, were expressed as relative-flux residuals with respect to the M4 field standard using $F/F_{\rm field} - 1$. Under this representation, the field standard defines the zero-residual baseline. The d/sd, sd, and esd spectra generally show positive residuals within $\mathcal{R}_{\rm YJ}$ and negative residuals within $\mathcal{R}_{\rm H}$ and $\mathcal{R}_{\rm K}$, whereas the field M4 test spectra remain closer to the baseline. This behavior motivated the use of field-referenced residual quantities as complementary spectral features.

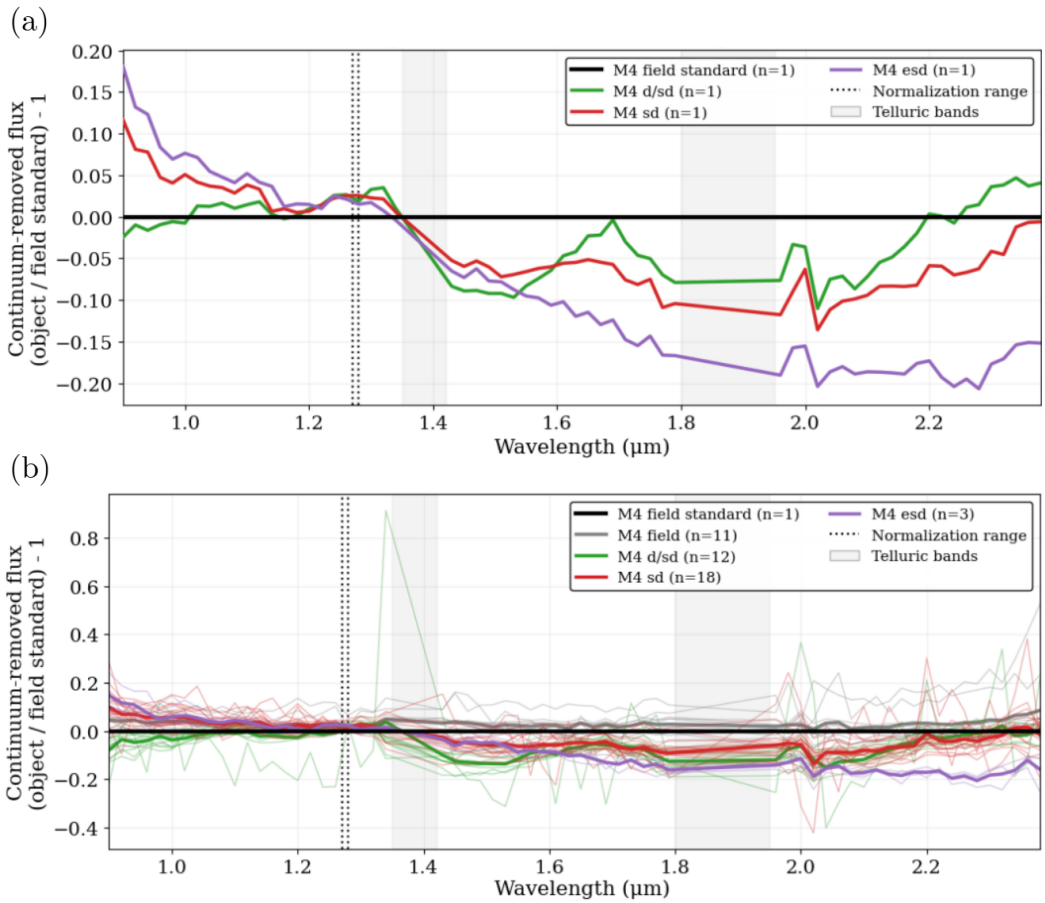


Figure 4: Illustrative field-referenced residual spectra for spectral type M4. (a) Residual spectra of the M4 d/sd, sd, and esd spectral standards computed relative to the M4 field spectral standard as $F/F_{\rm field} - 1$. (b) Residual spectra for M4 objects in the independent test set, including field objects shown in gray, computed relative to the same M4 field standard. In both panels, the black horizontal line marks the zero-residual field-reference baseline, the dotted vertical lines indicate the normalization interval, and the shaded regions mark the telluric absorption intervals.

To isolate peculiar-class-dependent spectral morphology from the broader variations associated with spectral type, each spectrum was expressed relative to the field spectral standard of the same spectral type. The field references were obtained exclusively from the spectral standards available in the training metadata file, thereby ensuring that the validation and test sets were not used to define the reference spectra. For each spectral type t, the corresponding reference spectrum was denoted as

$$F_{{\rm field},t}(\lambda_k), \tag{3}$$

where $F_{{\rm field},t}(\lambda_k)$ is the normalized flux of the unique field spectral standard associated with spectral type t at wavelength bin λ_k. Each spectrum was compared with the field standard sharing its spectral type, independently of its peculiar class. For example, an L6γ spectrum was compared with the L6 field spectral standard, whereas an sdM8 spectrum was compared with the M8 field spectral standard.

For each spectrum i, the relative-flux residual was calculated as

$$R_i(\lambda_k) = \frac{F_i(\lambda_k)}{F_{\mathrm{field},t_i}(\lambda_k)} - 1, \tag{4}$$

where $F_i(\lambda_k)$ is the normalized flux of spectrum i, and $F_{\mathrm{field},t_i}(\lambda_k)$ is the normalized flux of the field spectral standard corresponding to its spectral type t_i. Thus, $R_i(\lambda_k) = 0$ indicates agreement with the field standard of the same spectral type at wavelength bin λ_k, whereas positive and negative values indicate higher and lower relative flux, respectively.

A robust measure of the overall deviation from the field standard was defined as

$$A_i = \mathrm{median}_{\lambda_k \in \Lambda_i} \left| R_i(\lambda_k) \right|, \tag{5}$$

where Λ_i denotes the set of valid, unmasked wavelength bins for spectrum i. The dimensionless feature A_i represents the typical absolute relative-flux deviation from the field standard of the same spectral type, independently of the direction of the deviation.

To preserve the direction of the spectral differences, signed mean residuals were calculated over the $\mathcal{R}_{\mathrm{YJ}}$, $\mathcal{R}_{\mathrm{H}}$, and $\mathcal{R}_{\mathrm{K}}$ intervals:

$$\begin{aligned}
Y_i &= \frac{1}{N_{\mathrm{YJ},i}} \sum_{\lambda_k \in \Lambda_i \cap \mathcal{R}_{\mathrm{YJ}}} R_i(\lambda_k), \qquad \mathcal{R}_{\mathrm{YJ}} = [0.90, 1.34]\ \mu\mathrm{m}, \\
H_i &= \frac{1}{N_{\mathrm{H},i}} \sum_{\lambda_k \in \Lambda_i \cap \mathcal{R}_{\mathrm{H}}} R_i(\lambda_k), \qquad \mathcal{R}_{\mathrm{H}} = [1.43, 1.79]\ \mu\mathrm{m}, \\
K_i &= \frac{1}{N_{\mathrm{K},i}} \sum_{\lambda_k \in \Lambda_i \cap \mathcal{R}_{\mathrm{K}}} R_i(\lambda_k), \qquad \mathcal{R}_{\mathrm{K}} = [1.96, 2.38]\ \mu\mathrm{m}.
\end{aligned} \tag{6}$$

where $N_{\mathrm{YJ},i}$, $N_{\mathrm{H},i}$, and $N_{\mathrm{K},i}$ are the numbers of valid, unmasked wavelength bins within the corresponding regions. Positive values indicate that the spectrum lies above the field reference on average within a given region, whereas negative values indicate that it lies below the reference.

An additional contrast feature was defined to summarize the relative behavior of the $\mathcal{R}_{\mathrm{YJ}}$ region with respect to the $\mathcal{R}_{\mathrm{H}}$ and $\mathcal{R}_{\mathrm{K}}$ regions:

$$D_i = Y_i - \frac{H_i + K_i}{2}. \tag{7}$$

Positive values of D_i indicate that the mean residual in $\mathcal{R}_{\mathrm{YJ}}$ is greater than the average of the mean residuals in $\mathcal{R}_{\mathrm{H}}$ and $\mathcal{R}_{\mathrm{K}}$, whereas negative values indicate the opposite behavior.

The resulting field-referenced feature vector for each spectrum was therefore

$$\mathbf{z}_i = [A_i,\, Y_i,\, H_i,\, K_i,\, D_i]\,. \tag{8}$$

These five quantities jointly describe the magnitude and direction of the spectral deviation from the field standard of the same spectral type. The same procedure was applied to all spectra in the training, validation, and test sets for which a corresponding field spectral standard was available.

Figure 5 illustrates the residual-feature extraction procedure using a representative L6γ spectrum from the test set. Panel (a) compares the normalized object spectrum with the field standard of the same spectral type, while panel (b) shows the corresponding relative-flux residual spectrum used to calculate A_i and the signed mean residuals in $\mathcal{R}_{\mathrm{YJ}}$, $\mathcal{R}_{\mathrm{H}}$, and $\mathcal{R}_{\mathrm{K}}$. In panel (b), the horizontal zero level represents agreement with the field reference, and the green, purple, and orange segments indicate the regional mean residuals, respectively. The gray shaded intervals mark the masked telluric absorption regions, and the vertical dotted lines delimit the normalization interval.

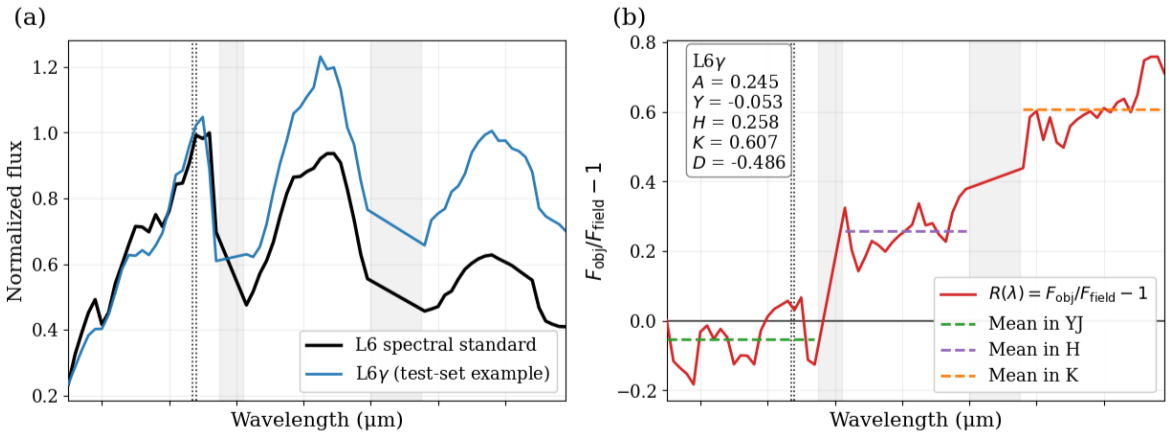


Figure 5: Field-referenced residual-feature extraction for a representative L6γ spectrum from the test set. (a) Normalized object and same-spectral-type field-standard spectra. (b) Relative-flux residual spectrum and regional mean residuals used to construct the features.

2.4 Photometric Data Retrieval and Missing-Data Imputation

As an initial step, the availability of the apparent J-band, H-band, and K-band magnitudes and the absolute J-band magnitude (M_J; `AbsMag`) was examined in the training, validation, and test sets. As shown in Fig. 6, only the apparent J- and H-band magnitudes were complete for the 66 retained spectral standards used to construct the synthetic training set. The apparent K-band magnitude and M_J were therefore excluded as photometric features because their incomplete coverage prevented their consistent incorporation across the training, validation, and test sets.

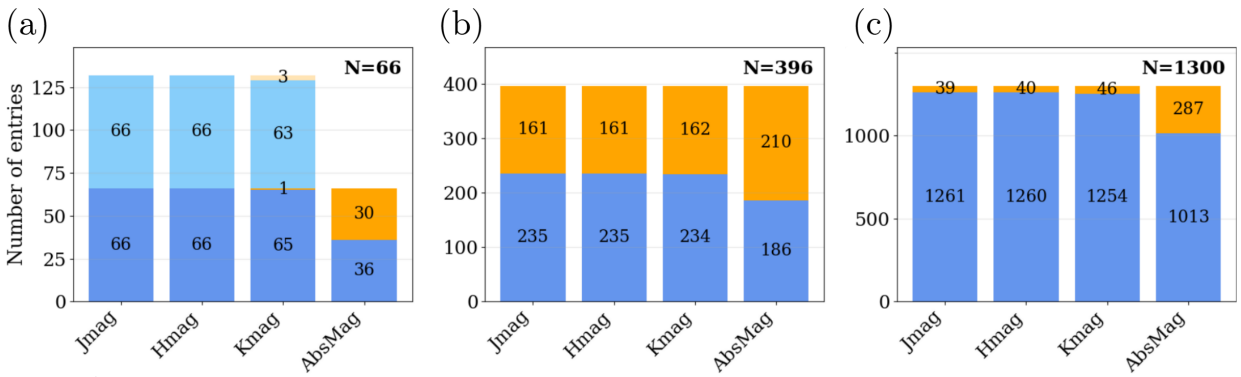


Figure 6: Availability of the apparent J-band, H-band, and K-band magnitudes and the absolute J-band magnitude in the (a) spectral standards used to generate the synthetic training spectra, (b) validation set, and (c) test set before photometric retrieval and imputation.

The apparent J- and H-band magnitudes were also incomplete in the validation and test samples. In the validation set, 161 entries lacked both magnitudes, corresponding to 148 synthetic spectra added by Zhou et al.[14] and 13 observed spectra. In the test set, 39 J-band magnitudes and 40 H-band magnitudes were missing. Consequently, synthetic and observed spectra were treated separately.

For synthetic spectra in the training and validation sets, each spectrum was associated with its parent spectral standard using the object name, spectral-type label, and peculiar-class label. The parent apparent J- and H-band magnitudes and their uncertainties were then obtained from the spectral-standard metadata file. New magnitudes were generated by sampling Gaussian distributions centered on the parent magnitudes, with the

corresponding photometric uncertainties adopted as standard deviations:

$$J_{\mathrm{syn},i} \sim \mathcal{N}\left(J_{\mathrm{parent}}, \sigma^2_{J,\mathrm{parent}}\right),$$
$$H_{\mathrm{syn},i} \sim \mathcal{N}\left(H_{\mathrm{parent}}, \sigma^2_{H,\mathrm{parent}}\right), \tag{9}$$

where $m_{J,\mathrm{parent}}$ and $m_{H,\mathrm{parent}}$ are the apparent magnitudes of the parent standard, and $\sigma_{J,\mathrm{parent}}$ and $\sigma_{H,\mathrm{parent}}$ are their corresponding photometric uncertainties. A fixed random seed was used to ensure reproducibility.

Observed validation and test spectra were processed using a two-stage procedure. First, missing J- and H-band measurements were searched directly in SPLAT.[19] For the test set, this search recovered 35 J-band measurements and 33 H-band measurements, leaving four J-band values and seven H-band values unavailable. Any remaining missing values in the observed validation and test samples were then imputed using the median magnitude of the available sources sharing the same spectral-type and peculiar-class labels. For a missing apparent magnitude $m_{i,b}$ in band $b \in \{J, H\}$, the imputed value was calculated as

$$\widehat{m}_{i,b} = \mathrm{median}\left\{m_{j,b} : s_j = s_i,\ d_j = d_i,\ m_{j,b}\ \mathrm{available}\right\}, \tag{10}$$

where s_i and c_i denote the spectral-type and peculiar-class labels of source i, respectively. The median was selected because it is less sensitive to extreme values than the mean and is more robust for categories represented by relatively few sources. Related statistical imputation approaches have previously been used to complete missing attributes in astronomical classification datasets.[29]

After applying the photometric retrieval and imputation procedures, the resulting m_J and m_H distributions were inspected across the training, validation, and test sets, as shown in Fig. 7. The three datasets show broadly overlapping magnitude ranges among the field, β, γ, d/sd, sd, and esd peculiar classes. The field class spans the widest range, particularly in the training and test sets, reflecting its larger sample size and broader photometric diversity. The remaining peculiar classes generally occupy similar magnitude intervals, although differences in their medians and dispersions are visible across the datasets. Therefore, m_J and m_H may provide complementary information for classification, but they are not sufficient by themselves to distinguish the peculiar classes.

2.5 Classification Models

Four supervised machine-learning models were evaluated for two classification outputs: spectral-type classification across the M0–T9 range and peculiar-class classification among the field, β, γ, d/sd, sd, and esd classes. Each model used the complete 90-dimensional input vector, comprising 67 normalized spectral bins, 16 targeted spectral slopes from Sec. 2.3.1, the apparent J- and H-band magnitudes from Sec. 2.4, and the five field-referenced residual features A_i, Y_i, H_i, K_i, and D_i from Sec. 2.3.2.

For direct comparison with Zhou et al.,[14] the RF, SVM, and KNN configurations reported in their study were included as reference candidates. Because the expanded feature representation changes the dimensionality and distribution of the input space, 30 additional configurations were randomly sampled from a predefined hyperparameter space for each classifier. The CNN was also evaluated to examine whether learned local spectral patterns could improve classification relative to the conventional models.

Each candidate configuration was trained on the synthetic training set and evaluated exclusively on the validation set. Peculiar-class BAcc was adopted as the primary selection criterion because the field class substantially outnumbers the β, γ, d/sd, sd, and esd classes. Peculiar-class Acc was also reported to facilitate direct comparison with Zhou et al.,[14] who evaluated model performance using accuracy. The independent test set was reserved exclusively for the final evaluation of the selected configurations.

The RF, SVM, and KNN classifiers were implemented using the `MultiOutputClassifier` module from `scikit-learn`, which fits a separate estimator for each classification output.[†30] The CNN was developed using TensorFlow and its Keras high-level API.[‡31,32] The models were trained using 66,000 synthetic spectra, selected using the 396-spectrum validation set, and assessed using the independent test set of 1,300 observed spectra.

[†] https://scikit-learn.org/stable/modules/generated/sklearn.multioutput.MultiOutputClassifier.html
[‡] https://www.tensorflow.org/ap?_docs/python/tf/keras

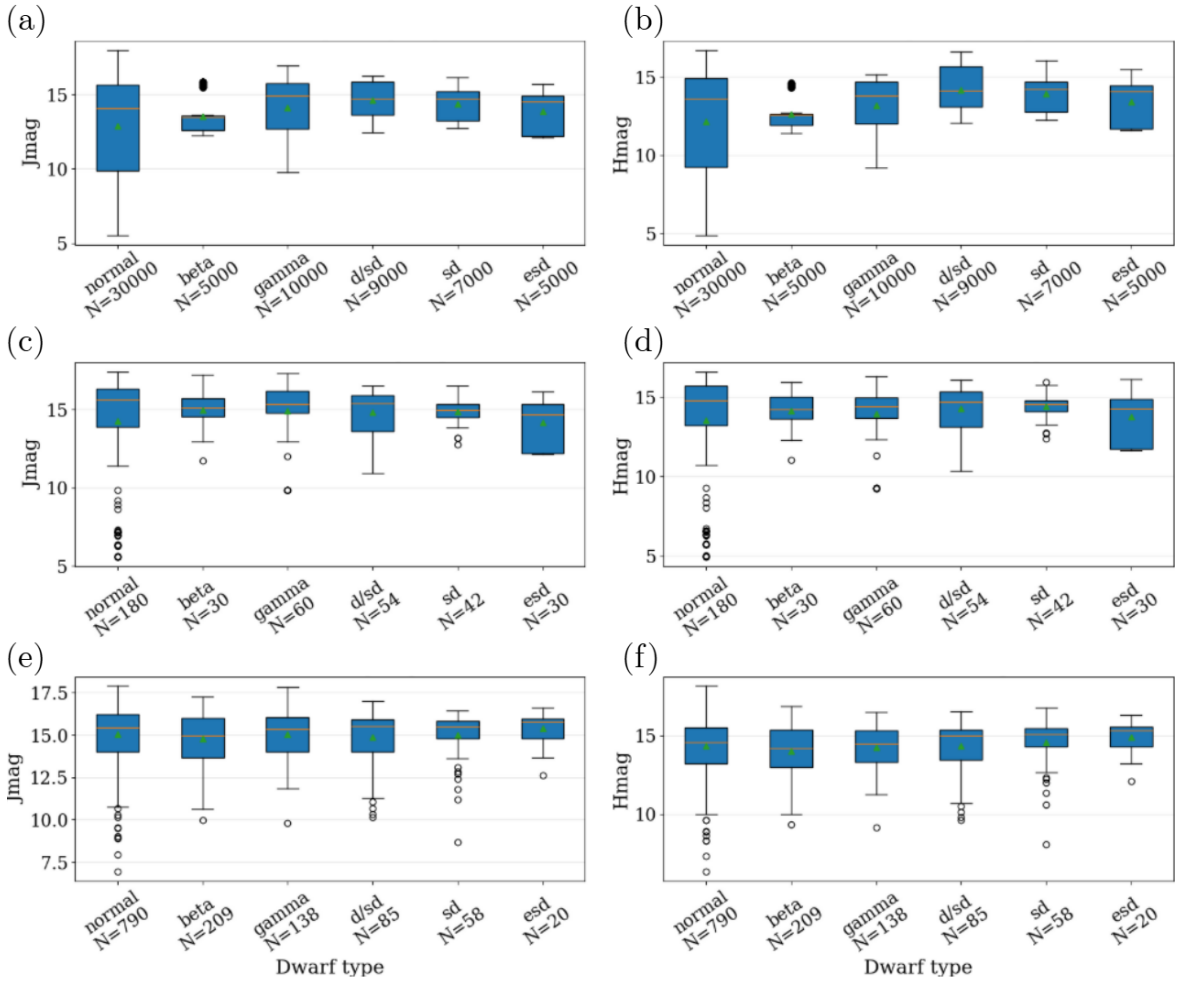


Figure 7: Post-imputation distributions of the apparent J- and H-band magnitudes (m_J and m_H) by peculiar class for the training, validation, and test sets. Panels (a) and (b) show the training set, panels (c) and (d) show the validation set, and panels (e) and (f) show the independent test set. Boxes indicate the interquartile range, orange horizontal lines indicate the medians, green triangles indicate the means, whiskers show the nonoutlier range, and open circles indicate outliers.

Before fitting the SVM, KNN, and CNN models, `StandardScaler` was fitted using only the training data and then applied without refitting to the validation and test sets. Standardization was used because SVM and KNN are sensitive to feature scale and because it stabilizes gradient-based optimization in the CNN. It was not applied to RF because decision-tree splits are invariant to monotonic feature scaling.

2.5.1 Random Forest

RF is an ensemble method in which multiple decision trees are trained using bootstrap samples and randomly selected feature subsets.[33] Their predictions are combined by majority voting, reducing the sensitivity of the classifier to noise and overfitting.

The reference configuration reported by Zhou et al.[14] used 700 trees (`n_estimators = 700`), no predefined maximum depth (`max_depth = None`), the square root of the total number of input features at each split (`max_features = sqrt`), a minimum of five samples to split an internal node (`min_samples_split = 5`), and a minimum of one sample per terminal leaf (`min_samples_leaf = 1`).

The additional RF candidates were sampled using `n_estimators` $\in \{300, 500, 700, 1000\}$, `max_depth` $\in \{20, 40, 60, \texttt{None}\}$, and `max_features` $\in \{\texttt{sqrt}, \texttt{log2}, 0.25, 0.50\}$. The minimum numbers of samples required

to split a node and occupy a terminal leaf were selected from $\{2, 5, 10\}$ and $\{1, 2, 4\}$, respectively. These combinations explored different ensemble sizes, tree complexities, and regularization levels. RF also enabled the relative importance of each feature to be estimated from its contribution to reducing node impurity. For a detailed description of the RF implementation adopted in the reference study, we refer the reader to Section 4.1.2 of Zhou et al.[14].

2.5.2 Support Vector Machine

SVM classifies observations by constructing a decision boundary that maximizes the margin between classes.[34] The samples located closest to this boundary are known as support vectors and determine the position of the separating hyperplane. When the classes cannot be separated linearly in the original feature space, a kernel function can be used to construct nonlinear decision boundaries.

The reference configuration reported by Zhou et al.[14] used $C = 10$, `kernel = rbf`, `gamma = scale`, and `decision_function_shape = ovo`. In this study, the regularization parameter was evaluated using

$$C \in \{0.1,\ 0.5,\ 1,\ 2,\ 5,\ 10,\ 20,\ 50,\ 100\},$$

and the kernel coefficient was sampled from

$$\gamma \in \{\texttt{scale},\ \texttt{auto},\ 10^{-4},\ 5\times10^{-4},\ 10^{-3},\ 5\times10^{-3},\ 10^{-2},\ 5\times10^{-2},\ 10^{-1}\}.$$

The RBF kernel and one-versus-one multiclass formulation were maintained for all configurations.

For `gamma = scale`, the kernel coefficient is calculated as

$$\gamma = \frac{1}{N_f \operatorname{Var}(X)},$$

where N_f is the number of input features and $\operatorname{Var}(X)$ is the variance of the training data after standardization. For `gamma = auto`, $\gamma = 1/N_f$. Smaller values produce smoother decision boundaries, whereas larger values increase the local influence of individual training samples.

2.5.3 k-Nearest Neighbors

KNN is a nonparametric classifier that assigns an observation to the class represented by its k nearest training samples.[35] For each input spectrum, the spectral-type and peculiar-class outputs were determined independently from the labels of the selected neighbors.

The reference configuration reported by Zhou et al.[14] used $k = 29$, uniform neighbor weighting, and the Minkowski distance with $p = 2$, equivalent to Euclidean distance. In this study, the number of neighbors was evaluated using

$$k \in \{3,\ 5,\ 7,\ 9,\ 11,\ 15,\ 21,\ 29,\ 35,\ 45,\ 55\}.$$

Uniform and distance-based weighting were considered, and the Minkowski parameter was selected from $p \in \{1, 2\}$.

For two feature vectors $\mathbf{x}_i$ and $\mathbf{x}_j$, the Minkowski distance is defined as

$$d(\mathbf{x}_i, \mathbf{x}_j) = \left[\sum_{m=1}^{N_f} |x_{i,m} - x_{j,m}|^p\right]^{1/p}, \tag{11}$$

where N_f is the total number of input features. The value $p = 1$ corresponds to the Manhattan distance, whereas $p = 2$ corresponds to the Euclidean distance. Smaller values of k preserve highly localized patterns but may increase sensitivity to noise, whereas larger values produce smoother decision boundaries by incorporating information from a broader neighborhood.

2.5.4 Convolutional Neural Network

The CNN was included to evaluate whether learned nonlinear representations could improve spectral-type and peculiar-class classification relative to the conventional models.[36] It received the same 90-dimensional input vector described above, arranged by concatenating the 67 spectral bins, 16 slopes, two photometric magnitudes, and five field-referenced residual features. Convolutional filters were applied along the complete vector, although only the spectral bins retain a continuous wavelength ordering. The network used a shared representation followed by two output heads: a 30-neuron softmax layer for spectral-type classification and a six-neuron softmax layer for peculiar-class classification.

The baseline architecture comprised three convolutional blocks with 32, 64, and 96 filters and kernel sizes of 5, 5, and 3, respectively. Each block used rectified linear unit activation, same padding, batch normalization, and L_2 regularization with a coefficient of 10^{-4}. Max pooling with a pool size of 2 was applied after the first two blocks, and dropout rates of 0.20, 0.25, and 0.35 were used after the three blocks. Global average pooling was followed by a fully connected layer with 64 neurons and a dropout rate of 0.40.

To explore different levels of network capacity and regularization, the numbers of filters in the three convolutional blocks were selected from $\{24, 32, 48\}$, $\{48, 64, 96\}$, and $\{64, 96, 128\}$, respectively. The first kernel size was selected from $\{3, 5, 7\}$, while the second and third were selected from $\{3, 5\}$. The corresponding dropout rates were evaluated over $\{0.15, 0.20, 0.25\}$, $\{0.20, 0.25, 0.30\}$, and $\{0.30, 0.35, 0.40\}$. The fully connected layer contained $\{48, 64, 96, 128\}$ neurons with a dropout rate selected from $\{0.30, 0.40, 0.50\}$. The L_2 coefficient was selected from $\{5 \times 10^{-5}, 10^{-4}, 2 \times 10^{-4}\}$, the initial learning rate from $\{10^{-4}, 3 \times 10^{-4}, 5 \times 10^{-4}, 10^{-3}\}$, and the batch size from $\{32, 64\}$. The three-block structure, batch-normalization layers, pooling operations, global average pooling, and two output heads were maintained across all configurations.

The baseline CNN was trained using the Adam optimizer[37] with an initial learning rate of 5×10^{-4}, a batch size of 32, and a maximum of 300 epochs. Sparse categorical cross-entropy was applied independently to both output heads. Early stopping monitored the combined validation loss with a patience of 12 epochs and a minimum improvement of 10^{-4}, restoring the weights associated with the lowest validation loss. In addition, `ReduceLROnPlateau` reduced the learning rate by a factor of 0.5 after five epochs without improvement, with a minimum learning rate of 10^{-6}. These training controls were maintained across all candidate configurations.

2.6 Evaluation Metrics

The performance of each classifier was evaluated independently for the spectral-type and peculiar-class outputs using Acc and BAcc. These metrics were obtained from the corresponding confusion matrices.[38,39] A confusion matrix summarizes the relationship between the true and predicted labels: its rows represent the true classes, its columns represent the predicted classes, the diagonal elements contain the correctly classified samples, and the off-diagonal elements show the specific class confusions produced by the model. Let C_{ij} denote the number of samples whose true class is i and whose predicted class is j. For a problem with N classes, the number of correctly classified samples is given by the sum of the diagonal elements, $\sum_{i=1}^{N} C_{ii}$.

For visualization, the confusion matrices were row-normalized so that each row sums to one. The normalized matrix was computed as

$$\widetilde{C}_{ij} = \frac{C_{ij}}{\sum_{j=1}^{N} C_{ij}}, \tag{12}$$

where $\widetilde{C}_{ij}$ represents the fraction of samples from true class i assigned to predicted class j. The diagonal entries $\widetilde{C}_{ii}$ therefore correspond to the recall of each class, whereas the off-diagonal entries indicate the distribution of its misclassified samples among the remaining classes.

Acc measures the proportion of correctly classified samples relative to the total number of evaluated samples

and is calculated as

$$\mathrm{Acc} = \frac{\sum_{i=1}^{N} C_{ii}}{\sum_{i=1}^{N}\sum_{j=1}^{N} C_{ij}}. \tag{13}$$

Acc provides a direct measure of overall predictive performance. However, it can be dominated by the most populated classes and may therefore provide an overly optimistic assessment when the class distribution is imbalanced.[39] This is particularly relevant for peculiar-class classification because the field class contains substantially more samples than the β, γ, d/sd, sd, and esd classes.

To account for this imbalance, BAcc was also calculated. BAcc corresponds to the unweighted mean of the recall obtained for each class, ensuring that every class contributes equally to the final score regardless of its number of samples.[39] For class i, recall is defined as the fraction of samples from that class that are correctly identified,

$$\mathrm{Recall}_i = \frac{TP_i}{TP_i + FN_i} = \frac{C_{ii}}{\sum_{j=1}^{N} C_{ij}}, \tag{14}$$

where TP_i denotes the correctly classified samples from class i, and FN_i denotes the samples belonging to class i that were assigned to another class. BAcc is then calculated as

$$\mathrm{BAcc} = \frac{1}{N}\sum_{i=1}^{N} \frac{C_{ii}}{\sum_{j=1}^{N} C_{ij}}. \tag{15}$$

Acc and BAcc were calculated separately for spectral-type and peculiar-class classification. Acc describes the overall fraction of correct predictions, whereas BAcc evaluates whether the classifier performs consistently across both well-represented and minority classes. Because the peculiar-class distribution is imbalanced, BAcc was prioritized as the primary model-selection metric.

For spectral-type classification, exact-match Acc was complemented by tolerance-based accuracies within ±1 and ±3 spectral types.[14]

Zhou et al.[14] reported conventional Acc values with binomial standard errors. Acc was therefore retained to permit direct comparison with their results. In contrast, peculiar-class BAcc cannot be compared strictly one-to-one with their Acc because the two metrics weight the classes differently. Nevertheless, the Acc values reported by Zhou et al. were used as reference benchmarks, and the present study sought BAcc values that met or exceeded those values while assigning equal importance to all peculiar classes. Because BAcc is computed as the mean of the class-specific recalls rather than as a single binomial proportion, the same binomial standard-error formulation was not applied. Peculiar-class BAcc was therefore reported as a point estimate and complemented by class-specific recalls and row-normalized confusion matrices.

3. RESULTS

RF, SVM, KNN, and CNN models were evaluated using the complete 90-dimensional input vector. Each model produced spectral-type predictions across 30 spectral types spanning M0–T9 and peculiar-class predictions among the field, β, γ, d/sd, sd, and esd classes. The models were trained using 66,000 synthetic spectra, while the validation and independent test sets contained 396 and 1,300 spectra, respectively. Model configurations were selected on the validation set using peculiar-class BAcc as the primary criterion and were subsequently evaluated on the independent test set.

The following subsections present the behavior of the field-referenced residual features, validation-based model selection, independent test-set performance, class-specific error patterns, and RF feature-importance analysis.

3.1 Behavior of Field-Referenced Residual Features

After the field-referenced feature-engineering procedure was completed, the resulting quantities were visually inspected to determine whether they captured structured differences among peculiar classes. This analysis was descriptive and was not used for hyperparameter selection, which was based exclusively on the validation metrics described in Sec. 2.5. Figures 8 and 9 provide complementary views of the field-referenced features for the representative spectral types M8 and L0. These spectral types include several peculiar classes and therefore allow the behavior of the features to be examined under different metallicity and surface-gravity regimes.

Figure 8 presents two-dimensional distributions of the contrast feature D_i and the global absolute-deviation feature A_i. The vertical coordinate A_i measures the overall magnitude of the spectral deviation from the field standard of the same spectral type, whereas the horizontal coordinate D_i represents the contrast between the mean residual in $\mathcal{R}_{\rm YJ}$ and the average of the mean residuals in $\mathcal{R}_{\rm H}$ and $\mathcal{R}_{\rm K}$. To reduce overplotting, a random 20% subsample of the synthetic training spectra from each peculiar class was displayed.

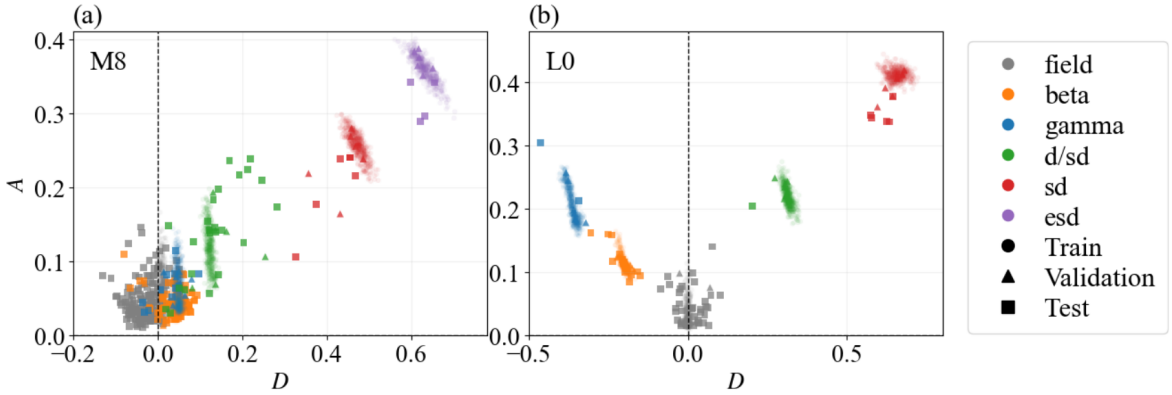


Figure 8: Distributions of the field-referenced features D_i and A_i for spectral types M8 and L0. Colors identify the peculiar classes, whereas marker shapes distinguish the training, validation, and test sets. To reduce overplotting, only a random 20% subsample of the synthetic training spectra from each peculiar class is shown.

Figure 9 shows the regional residual patterns underlying the contrast feature D_i. Each marker represents the median signed residual in $\mathcal{R}_{\rm YJ}$, $\mathcal{R}_{\rm H}$, or $\mathcal{R}_{\rm K}$ for a given peculiar class, and the error bars indicate the corresponding interquartile range. The columns distinguish the training, validation, and independent test sets. Together, Figs. 8 and 9 show both the distribution in the D_i–A_i feature space and the regional residual behavior producing that distribution.

The field-referenced features show different degrees of separation among peculiar classes depending on spectral type. For M8, the field, β, and γ classes overlap at low A_i values, whereas the sd and esd objects show larger deviations toward positive D_i and higher A_i. At this spectral type, the field-referenced features therefore provide stronger separation for the metal-poor classes than for the low-gravity classes. For L0, the distinction between low-gravity and metal-poor objects is more evident: the β and γ objects are preferentially located at negative D_i, whereas the d/sd and sd objects occupy positive D_i. The regional profiles indicate that this separation is driven by different signed residual patterns across $\mathcal{R}_{\rm YJ}$, $\mathcal{R}_{\rm H}$, and $\mathcal{R}_{\rm K}$.

To examine whether these trends persist beyond individual spectral types, the field-referenced features were visualized for the combined training, validation, and test sets in the A_i–D_i plane. Figure 10 shows this distribution using colors to identify peculiar classes and marker shapes to identify the M, L, and T spectral classes. Field objects are concentrated mainly at low A_i values and near $D_i = 0$, whereas the β and γ objects occur more frequently at negative D_i. In contrast, d/sd, sd, and esd objects are more commonly distributed toward positive D_i and higher A_i. Objects belonging to the T spectral class remain concentrated near the field locus because the available sample within this spectral class contains only field objects.

Although the A_i–D_i representation does not fully separate all peculiar classes, it reveals structured trends associated with spectral class and peculiar class, particularly between low-gravity and metal-poor objects. These

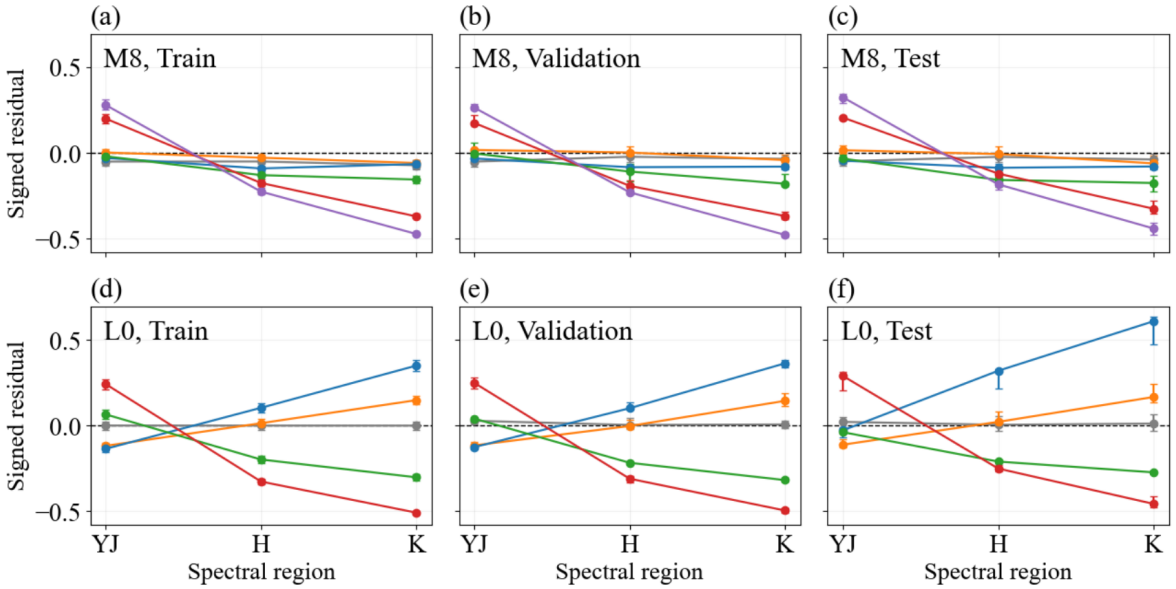


Figure 9: Median field-referenced residual profiles for spectral types M8 and L0. Colors identify the peculiar classes using the same color scheme as in Fig. 8. The horizontal dashed line marks zero residual relative to the field standard of the same spectral type.

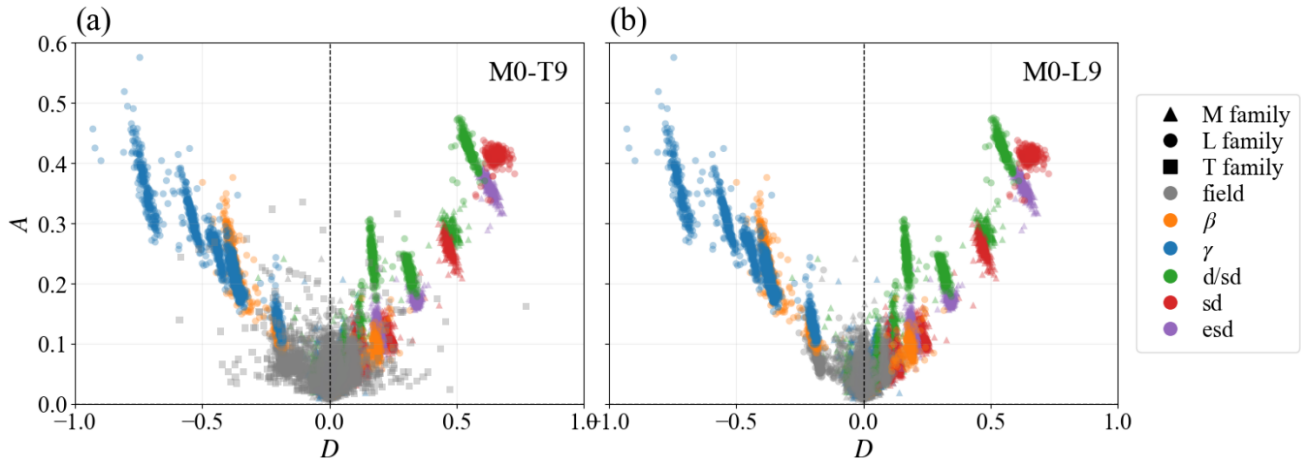


Figure 10: Distribution of the field-referenced features A_i and D_i for the combined training, validation, and test sets. Colors identify the peculiar classes using the same color scheme as in Fig. 8 and 9. A random 20% subsample of the synthetic training spectra is shown to reduce overplotting. The left panel shows the complete M0–T9 spectral-type range, whereas the right panel is restricted to the M0–L9 range by excluding the T spectral class.

results support the inclusion of the five field-referenced features A_i, Y_i, H_i, K_i, and D_i as complementary inputs to the machine-learning classifiers.

3.2 Validation-Based Model Selection

One configuration was selected for each classifier based on the highest peculiar-class BAcc obtained on the validation set.

Table 1 compares the selected configurations with the reference configurations reported by Zhou et al.[14] For RF, SVM, and KNN, the reference configurations were retrained using the present 90-dimensional input vector and evaluated on the validation set used in this work. Thus, the reference values shown in the table were obtained in the present analysis rather than taken directly from the published results of Zhou et al. The table reports Acc

and BAcc for peculiar-class and spectral-type classification. Spectral-type accuracies within ±1 and ±3 spectral types are reported only for the independent test set because they were not used for model selection.

Table 1: Validation-set performance of the reference and selected model configurations.

Model	Configuration	Peculiar BAcc	Peculiar Acc	SpT BAcc	SpT Acc
RF	Selected	91.42%	90.66%	88.29%	86.62%
RF	Reference	89.48%	89.14%	86.45%	83.59%
SVM	Selected	87.10%	86.62%	82.51%	85.10%
SVM	Reference	83.21%	82.58%	92.18%	**93.69%**
KNN	Selected	**94.88%**	**94.19%**	**92.36%**	90.91%
KNN	Reference	90.44%	88.64%	84.92%	84.09%
CNN	Selected	88.18%	86.36%	76.51%	76.26%

Note. Reference denotes the configuration reported by Zhou et al., retrained using the present feature representation and evaluated on the validation set used in this work. These values were obtained in the present analysis. The highest value in each metric column is indicated in bold.

The hyperparameters of the configurations selected using peculiar-class BAcc are summarized in Table 2.

Table 2: Hyperparameter configurations selected using the validation set.

RF	**SVM**	**KNN**	**CNN**
`n_estimators`: 300	$C = 1$	`n_neighbors`: 3	Filters: $(32, 48, 96)$
`max_depth`: 60	$\gamma = 0.1$	`weights`: distance	Kernels: $(3, 3, 5)$
`max_features`: log2	`kernel`: rbf	`metric`: minkowski	Dropout: $(0.15, 0.25, 0.35)$
`min_samples_split`: 5	`decision_function_shape`:	$p = 1$	Dense units: 128
`min_samples_leaf`: 1	ovo		Dense dropout: 0.40
			$L_2 = 10^{-4}$
			Learning rate: 10^{-3}
			Batch size: 32

Among the selected configurations, KNN achieved the highest validation peculiar-class BAcc (94.88%), peculiar-class Acc (94.19%), and spectral-type BAcc (92.36%). RF improved all four validation metrics relative to its reference configuration. The selected SVM configuration improved both peculiar-class metrics but produced lower spectral-type Acc and BAcc than its reference configuration. CNN obtained lower validation performance than RF and KNN for both classification outputs.

3.3 Independent Test-Set Performance

The selected RF, SVM, KNN, and CNN models were evaluated on the independent test set to assess their performance on previously unseen observed spectra. Table 3 reports peculiar-class Acc and BAcc, together with spectral-type accuracies within ±1 and ±3 spectral types.

The RF, SVM, and KNN results obtained in this work are compared with the corresponding test-set metrics reported in Table 6 of Ref. 14. Unlike the validation reference values in Table 1, which were obtained by retraining the configurations of Zhou et al. using the present input representation, the values attributed to Zhou et al. in Table 3 were taken directly from their published results. CNN was not evaluated in the reference study.

Among the models evaluated in this work, KNN achieved the highest peculiar-class BAcc (89.84%) and Acc (88.08%). It also obtained the highest spectral-type accuracies within ±1 and ±3 spectral types, reaching 96.85% and 99.54%, respectively. Relative to the published KNN results of Zhou et al., the spectral-type accuracies increased by 1.39 and 0.62 percentage points, whereas peculiar-class Acc was 1.38 percentage points lower.

The row-normalized spectral-type confusion matrices for the selected RF, SVM, KNN, and CNN classifiers are provided in Appendix B. These matrices complement the tolerance-based accuracies by showing the exact spectral-type recall and the distribution of predictions among neighboring and more distant spectral types.

Table 3: Independent test-set performance of the RF, SVM, KNN, and CNN classifiers and comparison with the results reported by Zhou et al.

Model	Source	Peculiar-Class BAcc	Peculiar-Class Acc	SpT Acc (±1 Subtype)	SpT Acc (±3 Subtypes)
RF	This work	84.35%	81.85%	92.08%	99.08%
RF	Zhou et al.	–	86.15%	91.54%	98.15%
SVM	This work	77.78%	72.38%	89.92%	95.77%
SVM	Zhou et al.	–	82.69%	91.08%	98.92%
KNN	This work	**89.84%**	88.08%	**96.85%**	**99.54%**
KNN	Zhou et al.	–	**89.46%**	95.46%	98.92%
CNN	This work	75.90%	68.62%	90.15%	98.69%

Note. The values attributed to Zhou et al. were taken directly from Table 6 of Ref. 14. The highest reported value in each metric column is indicated in bold. Metrics not reported by Zhou et al. are indicated by a dash.

Figure 11 summarizes the validation and independent test performance of the evaluated classifiers. Panel (a) compares the selected configurations with the reference configurations of Zhou et al. retrained in the present analysis, whereas panel (b) compares the independent test-set results with the published values of Zhou et al. Because the reference study did not report test-set peculiar-class BAcc, its curves connect only the available metrics.

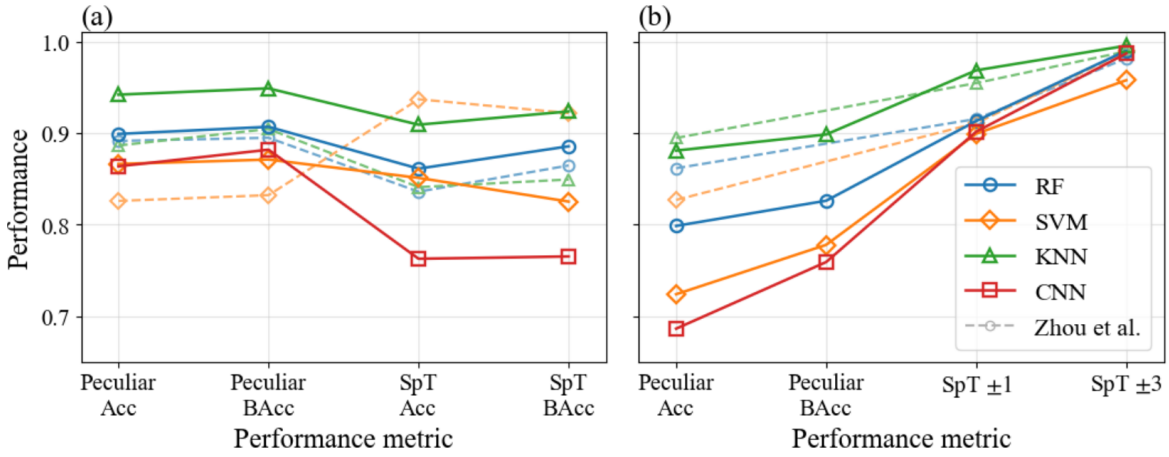


Figure 11: Validation and independent test performance of the evaluated classifiers. (a) Peculiar-class Acc, peculiar-class BAcc, spectral-type Acc, and spectral-type BAcc for the selected configurations and the reference configurations retrained in this work. (b) Independent test-set peculiar-class Acc, peculiar-class BAcc, and spectral-type accuracies within ±1 and ±3 spectral types, compared with the published results of Zhou et al.

Relative to validation, peculiar-class BAcc decreased by 7.07 percentage points for RF, 9.32 percentage points for SVM, 5.04 percentage points for KNN, and 12.28 percentage points for CNN. The corresponding decreases in peculiar-class Acc were 8.81, 14.24, 6.11, and 17.74 percentage points, respectively. KNN therefore exhibited the smallest validation–test performance gap, whereas CNN showed the largest reduction.

3.4 Class-Wise Performance and Error Analysis

To examine the class-specific behavior of the selected classifiers, row-normalized peculiar-class confusion matrices were computed on the independent test set for RF, SVM, KNN, and CNN, as shown in Fig. 12. Each row represents a true peculiar class and is normalized by the number of objects belonging to that class. Following the recall definition introduced in Sec. 2.6, the diagonal entries correspond to class-specific recall, whereas the off-diagonal entries show the predicted classes assigned to misclassified objects. The corresponding row-normalized spectral-type confusion matrices are provided in Appendix B.

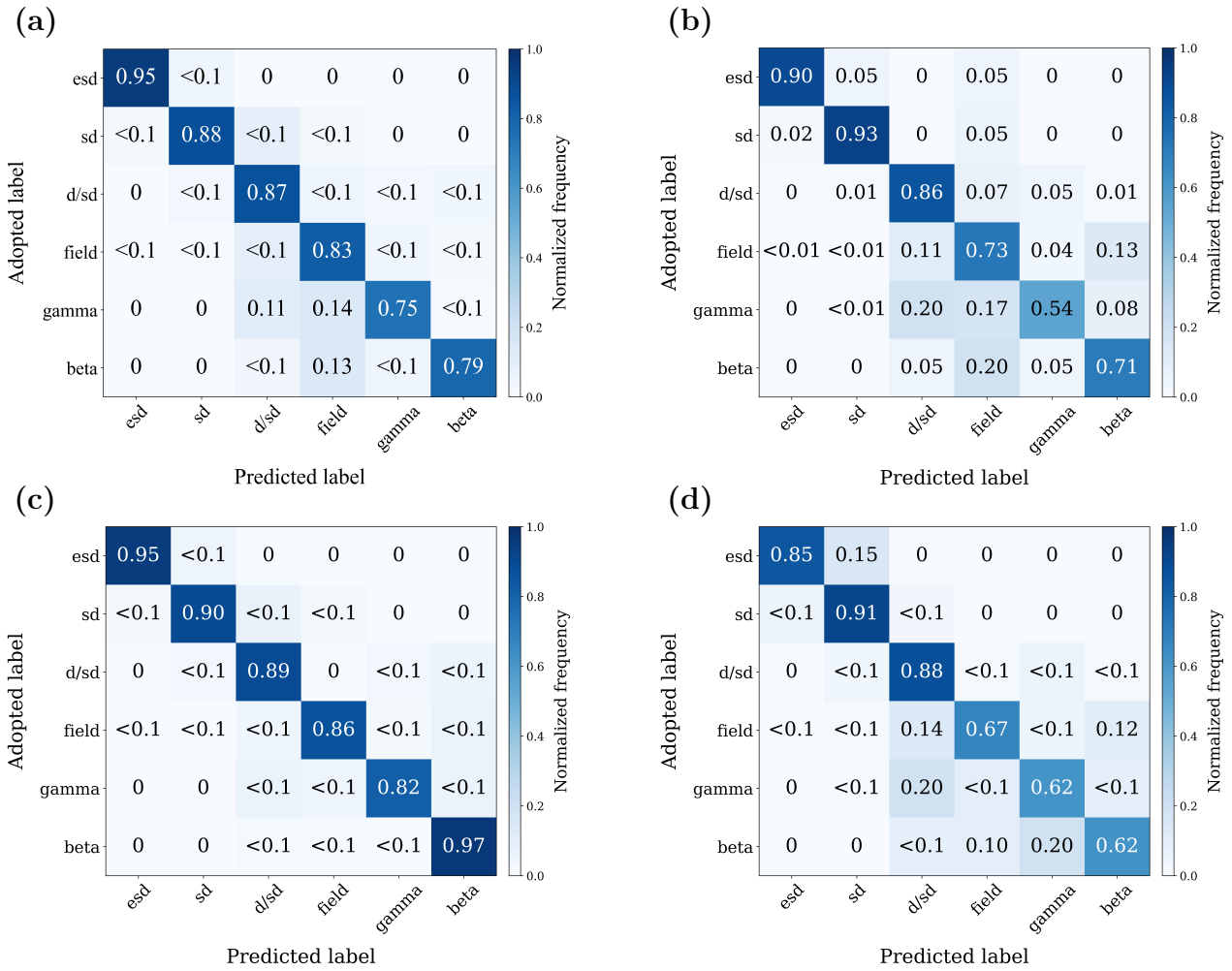


Figure 12: Row-normalized peculiar-class confusion matrices for the selected (a) RF, (b) SVM, (c) KNN, and (d) CNN models on the independent test set. Diagonal entries represent class-specific recall, while off-diagonal entries show the distribution of misclassifications.

KNN exhibited the most uniform diagonal, with class-specific recalls ranging from 0.82 to 0.97. It correctly classified 97% of the β objects, 95% of the esd objects, 90% of the sd objects, 89% of the d/sd objects, 86% of the field objects, and 82% of the γ objects. None of its off-diagonal entries exceeded 0.10. This performance is consistent with its peculiar-class BAcc of 89.84%, the highest obtained among the four classifiers.

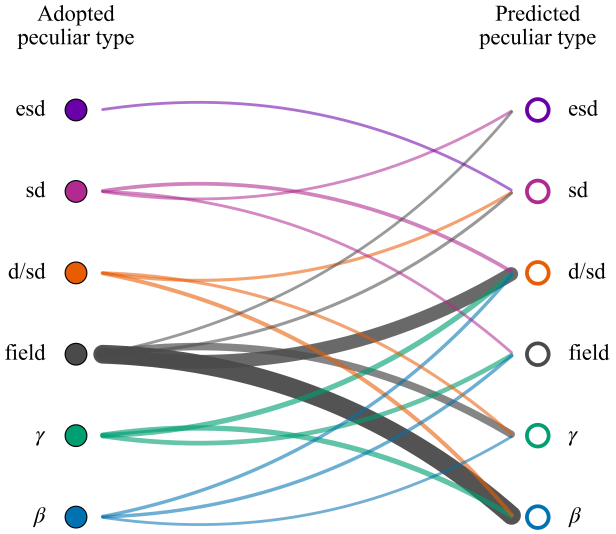
RF also maintained relatively high recall for esd (0.95), sd (0.88), d/sd (0.87), and field (0.83), whereas recall decreased to 0.79 for β and 0.75 for γ. Its largest confusions involved γ objects classified as field (0.14) or d/sd (0.11), and β objects classified as field (0.13).

SVM showed greater variation among peculiar classes. It achieved recalls of 0.90, 0.93, and 0.86 for esd, sd, and d/sd, respectively, but recall decreased to 0.73 for field, 0.71 for β, and 0.54 for γ. Its dominant errors included γ objects classified as d/sd (0.20) or field (0.17), and β objects classified as field (0.20).

CNN achieved relatively high recall for sd (0.91) and d/sd (0.88), but lower recall for field (0.67), β (0.62), and γ (0.62). Its main confusions involved γ objects classified as d/sd (0.20), β objects classified as γ (0.20), and field objects classified as d/sd (0.14) or β (0.12).

Across the four classifiers, the esd, sd, and d/sd classes were generally recognized more consistently than the field, β, and γ classes. KNN reduced the differences in recall among peculiar classes most effectively, whereas SVM and CNN exhibited less uniform diagonals and lower peculiar-class BAcc.

To further characterize the errors of the best-performing classifier, the peculiar-class misclassification flows of the selected KNN model were examined. Figure 13 shows the true–predicted pairs associated only with incorrectly classified objects.



Misclassified flows

true	→ pred	count	(% errors)
field	→ β	52	(33.5%)
field	→ d/sd	36	(23.2%)
field	→ γ	16	(10.3%)
γ	→ β	9	(5.8%)
γ	→ d/sd	9	(5.8%)
γ	→ field	7	(4.5%)
d/sd	→ β	5	(3.2%)
sd	→ d/sd	4	(2.6%)
β	→ d/sd	3	(1.9%)
β	→ field	3	(1.9%)
d/sd	→ γ	2	(1.3%)
d/sd	→ sd	2	(1.3%)
field	→ sd	2	(1.3%)
sd	→ esd	1	(0.6%)
field	→ esd	1	(0.6%)
esd	→ sd	1	(0.6%)
β	→ γ	1	(0.6%)
sd	→ field	1	(0.6%)

Figure 13: Peculiar-class misclassification flows for the selected KNN model on the independent test set. Line thickness is proportional to the number of objects in each true–predicted flow. The legend lists the corresponding counts and percentages relative to the total number of peculiar-class misclassifications.

The largest KNN misclassification flow corresponded to field objects classified as β, accounting for 33.5% of all peculiar-class errors. Field objects classified as d/sd represented a further 23.2% of the errors. Together, misclassified field objects accounted for approximately 69.0% of the total KNN peculiar-class misclassifications. Smaller flows occurred among the γ, β, and d/sd classes.

The spectral-type behavior of KNN was also examined as a function of the SNR values available in the test-set metadata. Following the SNR intervals adopted by Zhou et al.,[14] Fig. 14 shows the difference between the adopted and predicted spectral-type labels. An error of zero indicates an exact prediction, whereas nonzero values indicate deviations from the adopted spectral type.

The spectral-type errors were concentrated near zero across the evaluated SNR intervals, consistent with the high accuracies within ±1 and ±3 spectral types obtained by KNN. Nonzero errors remained present in both lower- and higher-SNR groups and were therefore not restricted to spectra within the lowest-SNR intervals.

3.5 RF Feature Importance

The RF classifier provides an impurity-based estimate of the normalized importance of each input feature. Figure 15 shows the 15 highest-ranked features for peculiar-class classification.

Among all input features, the field-referenced contrast feature D_i exhibited the highest normalized importance, 6.84%, followed by the global absolute-residual feature A_i, with 4.86%. The apparent J- and H-band magnitudes, m_J and m_H, ranked third and fourth, with importance values of 4.53% and 4.41%, respectively. The field-referenced regional feature K_i also appeared among the most influential inputs, with an importance of 3.07%.

Spectral slopes calculated over the 0.90–1.34 μm, 1.43–1.69 μm, 2.08–2.38 μm, and 1.43–1.585 μm intervals were also among the highest-ranked predictors, with importance values ranging from 1.78% to 2.79%. In addition, several individual spectral bins between approximately 0.89 and 0.99 μm were selected, with importance values ranging from 1.76% to 2.43%.

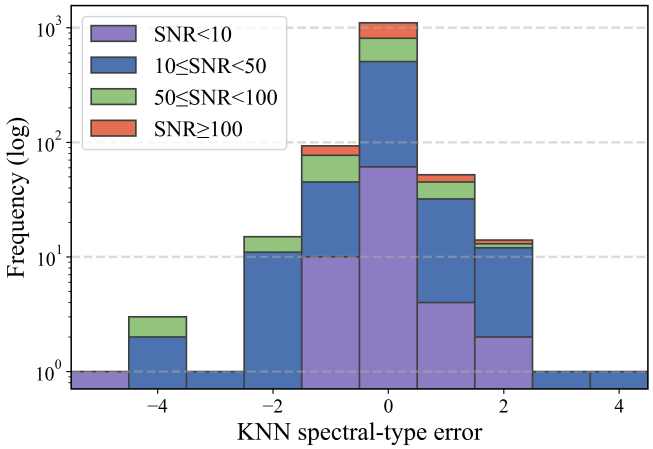


Figure 14: SNR-dependent spectral-type classification errors of the selected KNN model on the independent test set. Errors were calculated as the adopted spectral-type label minus the predicted label using the integer encoding M0 = 0, L0 = 10, and T0 = 20. The vertical axis shows logarithmic frequency, and the distributions are grouped according to the SNR intervals adopted by Zhou et al.[14]

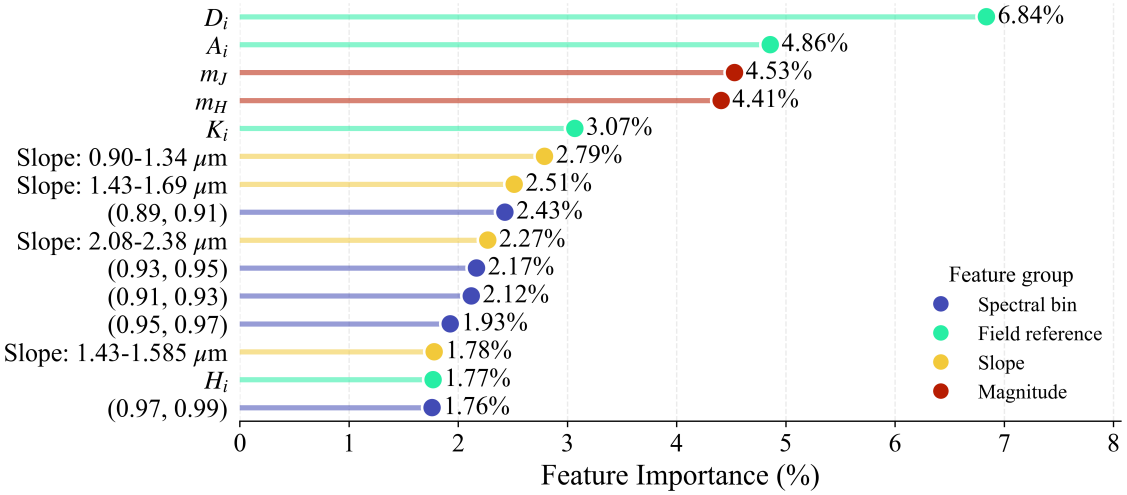


Figure 15: Top 15 input features ranked by normalized impurity-based importance for RF peculiar-class classification.

4. DISCUSSION

The present study examined whether an expanded feature representation and peculiar-class-oriented model selection could improve the simultaneous classification of spectral type and peculiar class. Peculiar-class BAcc was prioritized because it assigns equal weight to the recall of the field, β, γ, d/sd, sd, and esd classes, preventing the more prevalent field class from dominating model selection.

4.1 Effect of Peculiar-Class-Oriented Model Selection

The selected RF configuration improved all four validation metrics relative to the retrained reference configuration, suggesting that its more constrained tree depth and logarithmic feature sampling were better suited to the expanded feature representation on the validation set. The selected model also used fewer trees than the reference configuration, indicating that the improvement was associated with the hyperparameter combination rather than with a larger ensemble.

For SVM, prioritizing peculiar-class BAcc produced a clear trade-off between the two classification tasks. The selected configuration improved both peculiar-class metrics but reduced spectral-type Acc and BAcc relative to

the reference configuration. This behavior is consistent with the model-selection criterion, which favored balanced recognition of the six peculiar classes rather than maximizing spectral-type performance.

KNN exhibited the most consistent validation improvement. Reducing the neighborhood size from $k = 29$ to $k = 3$ allowed the classifier to preserve more localized structure and reduced the influence of more distant samples. Distance-weighted voting further increased the contribution of the closest spectra. In addition, the Manhattan distance may have reduced the influence of large differences in individual features within the heterogeneous 90-dimensional input space. Together, these changes improved both peculiar-class and spectral-type validation performance.

The CNN did not provide an advantage over RF or KNN despite its greater model complexity. One possible limitation is the heterogeneous organization of the input vector. The spectral bins follow a physically meaningful wavelength sequence, whereas the slopes, apparent magnitudes, and field-referenced features do not share the same local ordering. Applying one-dimensional convolutional filters to the complete concatenated representation may therefore combine adjacent features that are not physically related. In contrast, the conventional classifiers can use these engineered features without assuming that neighboring positions in the input vector represent local spectral structure.

4.2 Generalization and Validation–Test Shift

KNN provided the strongest class-balanced performance on the independent test set and exhibited the smallest validation–test gap among the four classifiers. Because Zhou et al. did not report peculiar-class BAcc, a direct comparison of class-balanced test performance was not possible. [14] For the directly comparable peculiar-class Acc metric, the selected KNN model remained 1.38 percentage points below the published reference value. However, it exceeded the published spectral-type accuracies within both ± 1 and ± 3 spectral types.

The smaller validation–test reduction obtained by KNN suggests that its local, distance-weighted decision rule transferred more consistently to the independent observed sample. By contrast, the larger decreases obtained by SVM and CNN may indicate greater sensitivity to characteristics specific to the validation distribution. Nevertheless, the spectral-type tolerance metrics remained high, particularly within ± 3 spectral types, indicating that most spectral-type errors remained close to the adopted classification.

The validation–test differences are consistent with a distribution mismatch between model development and final evaluation. The training set consisted entirely of synthetic spectra generated from spectral standards, and the validation set contained both observed and synthetic spectra, whereas the independent test set contained only observed spectra. The validation set may therefore retain greater similarity to the synthetic training distribution. In contrast, the larger and fully observed test set may contain a broader range of spectral quality, intrinsic variability, and peculiar-class diversity.

The KNN error flows were dominated by field objects assigned to non-field classes. Because field objects constitute the largest test-set class, the absolute flow percentages are influenced by class prevalence and should be interpreted together with the row-normalized confusion matrix. The remaining confusions among field, d/sd, γ, and β may reflect with overlap among the spectral characteristics associated with these peculiar classes.[6–10]

The presence of nonzero spectral-type errors across several SNR intervals indicates that low spectral quality alone does not explain the KNN misclassifications. This behavior is consistent with the findings of Zhou et al., who reported that higher-SNR spectra were not necessarily classified more accurately.[14] However, because the plotted frequencies were not normalized by the number of spectra in each SNR interval, the analysis does not directly quantify an SNR-dependent error rate. Instead, the figure shows that deviations from the adopted spectral type persist across the evaluated SNR range.

Finally, the selected configurations may reflect characteristics specific to the validation set because multiple candidates were evaluated using the same samples. This does not affect the independence of the final test evaluation, but it limits the extent to which the validation improvements can be generalized. Repeated stratified partitions or nested model-selection procedures could provide a more robust assessment while preserving a fully independent observed test set for final evaluation.

4.3 Implications for Peculiar-Class Classification and Feature Interpretation

Prioritizing peculiar-class BAcc changed the configurations selected for all three conventional classifiers and improved their class-balanced validation performance. Among the evaluated models, KNN provided the most favorable compromise between peculiar-class recognition and tolerance-based spectral-type accuracy. Its performance suggests that the engineered spectral, photometric, and field-referenced features form informative local neighborhoods when closer samples receive greater weight.

The objective of improving peculiar-class classification was partially achieved. The selected KNN model attained the highest peculiar-class BAcc among the four models evaluated in this work and improved the spectral-type accuracies within ±1 and ±3 spectral types relative to Zhou et al.[14] Nevertheless, its independent test-set peculiar-class Acc remained slightly below the published reference value. The main contribution is therefore not an improvement in every reported metric, but a model-selection framework that explicitly accounts for class imbalance and identifies a classifier with strong class-balanced performance.

Although KNN achieved the strongest overall peculiar-class performance, RF provided an additional interpretability advantage through its feature-importance estimates. The prominence of the field-referenced features D_i and A_i indicates that these variables summarize broad spectral properties useful for separating peculiar classes. Similarly, the high ranking of the apparent J- and H-band magnitudes, indicates that near-infrared photometric information provides discriminatory information beyond that contained in the normalized spectra.

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The spectral contributions identified in this work are broadly consistent with the feature-importance analysis of Zhou et al.[14] Their RF model assigned high importance to spectral features near 0.89 μm and 1.0 μm, and also identified relevant contributions near the 1.4 μm H_2O absorption band. Similarly, several of the highest-ranked spectral bins in the present analysis are located between approximately 0.89 and 0.99 μm. This short-wavelength region contains TiO- and FeH-sensitive spectral structure associated with variations in effective temperature, metallicity, and surface gravity.[6,9–11,40]

Zhou et al.[14] also found that the K-band region exhibited slightly greater importance than the H-band region and emphasized the overall contribution of the Z- and Y-band regions. In the present work, slopes spanning the H- and K-band regions were among the most influential features, particularly those calculated over 1.43–1.69 μm and 2.08–2.38 μm. The spectral morphology of these regions is known to respond to gravity- and composition-sensitive molecular absorption and continuum structure.[6,11,26] Both analyses therefore indicate that peculiar-class discrimination depends on information distributed across multiple near-infrared regions rather than on a single absorption feature. The simultaneous selection of broad slopes and individual spectral bins further suggests that continuum variations and localized absorption structures provide complementary discriminatory information.

A direct quantitative comparison of the feature-importance values must be made cautiously because the two studies use different input representations. Zhou et al. primarily evaluated normalized spectral fluxes, whereas the present analysis also incorporates engineered slopes, apparent photometric magnitudes, and field-referenced features. Zhou et al. further noted that the importance of spectral bins within their J-band normalization window was suppressed because normalization reduced the flux variance in that region. In contrast, m_J is an independent photometric measurement and should not be interpreted as equivalent to the normalized spectral bins within the J-band region.

Finally, the reported RF rankings correspond to impurity-based feature importance.[41] These values quantify the relative use of each input feature by the trained RF model but do not directly measure astrophysical significance or causal relevance. Impurity-based importance can be affected by predictor characteristics, while

correlations among features can alter or redistribute their apparent importance.[42,43] Therefore, the rankings should be interpreted as model-dependent evidence of discriminatory utility.

5. SUMMARY, CONCLUSIONS, AND FUTURE WORK

In this study, we investigated whether an expanded feature representation and a peculiar-class-oriented model-selection strategy could improve the simultaneous classification of spectral type and peculiar class for cool dwarfs spanning the M, L, and T spectral classes using low-resolution near-infrared spectra. Following the general framework of Zhou et al.,[14] RF, SVM, KNN, and an additional CNN model were evaluated using SpeX prism spectra. Unlike the reference work, which primarily emphasized spectral-type classification using binned mean fluxes, the present analysis prioritized peculiar-class BAcc during validation to account for the imbalanced distribution of field, β, γ, d/sd, sd, and esd objects.

The training set consisted of 66,000 synthetic spectra generated from the spectral standards after removing spectral types not represented in the validation or test sets. The validation and independent test sets contained 396 and 1,300 spectra, respectively. Each model used a 90-dimensional input vector comprising 67 binned spectral fluxes, 16 targeted spectral slopes, the apparent J- and H-band magnitudes, and five field-referenced residual features. The main findings are summarized below.

1. **Model performance.** Among the evaluated classifiers, KNN achieved the strongest overall performance. On the independent test set, KNN reached a peculiar-class BAcc of 89.84%, a peculiar-class Acc of 88.08%, and spectral-type accuracies of 96.85% and 99.54% within ±1 and ±3 spectral types, respectively. Compared with Zhou et al.,[14] the selected KNN model improved the spectral-type tolerance metrics by 1.39 percentage points within ±1 and 0.62 percentage points within ±3 spectral types. However, its peculiar-class Acc remained 1.38 percentage points below the value reported by Zhou et al. Because peculiar-class BAcc was not reported in the reference work, a direct comparison of class-balanced peculiar-class performance was not possible.

2. **Effect of peculiar-class-oriented model selection.** Prioritizing peculiar-class BAcc changed the selected hyperparameter configurations for RF, SVM, and KNN relative to the reference configurations. The clearest improvement was obtained with KNN, for which validation peculiar-class BAcc increased from 90.44% to 94.88%, while spectral-type BAcc increased from 84.92% to 92.36%. These results suggest that the expanded feature representation, combined with a smaller neighborhood size, distance weighting, and Manhattan distance, better preserved the local class structure than the reference KNN configuration.

3. **Generalization to the independent test set.** The validation gains did not transfer uniformly to the independent observed test set. All four classifiers showed lower peculiar-class performance on the test set than on the validation set, with KNN exhibiting the smallest validation–test reduction and CNN the largest. This behavior is consistent with a mismatch among the synthetic training distribution, the mixed validation set, and the fully observed independent test set. Nevertheless, the high KNN spectral-type accuracies within ±1 and ±3 spectral types indicate that most predictions remained close to the adopted spectral types, even when peculiar-class classification was more strongly affected by the distribution shift.

4. **Peculiar-class classification behavior.** The row-normalized confusion matrices showed that KNN provided the most uniform class-specific performance, with recalls ranging from 0.82 to 0.97 across the six peculiar classes. The misclassification-flow analysis showed that most KNN peculiar-class errors originated from field objects assigned to non-field classes. The dominant confusion was field objects classified as β, accounting for 33.5% of all KNN peculiar-class misclassifications, followed by field objects classified as d/sd, which accounted for 23.2%. These results indicate that the main remaining difficulty is not the overall separation of all peculiar classes, but the boundary between field objects and low-gravity or transitional peculiar classes.

5. **Feature importance.** The RF feature-importance analysis showed that the most influential variables for peculiar-class classification were not limited to individual spectral bins. The field-referenced features D_i and A_i ranked first and second, with normalized importance values of 6.84% and 4.86%, respectively. The

apparent J- and H-band magnitudes, m_J and m_H, also ranked among the four most important features. In addition, several spectral slopes and individual bins between approximately 0.89 and 0.99 μm contributed to the classification. This result is broadly consistent with Zhou et al.,[14] who identified high-importance spectral regions near 0.89 μm and 1.0 μm, associated with effective-temperature-, surface-gravity-, and metallicity-sensitive features such as TiO and FeH.

6. **SNR-dependent behavior.** The SNR analysis of the selected KNN model showed that spectral-type errors were concentrated near zero across the evaluated SNR intervals. However, nonzero errors were also present among higher-SNR spectra. This behavior agrees with the independent test-set analysis of Zhou et al.,[14] in which higher SNR did not necessarily result in more accurate classification.

Overall, the objective of improving peculiar-class classification was partially achieved. The selected KNN model produced the strongest class-balanced peculiar-class performance among the models evaluated in this work and improved the spectral-type tolerance metrics relative to Zhou et al.[14] Nevertheless, its overall peculiar-class Acc remained slightly below the published reference value. The main contribution of this work is therefore not an improvement in every reported metric, but the development of a feature-engineering and model-selection framework that explicitly accounts for peculiar-class imbalance and provides a more detailed assessment of class-specific performance.

Future work should expand the set of spectral standards, particularly for sparsely represented low-gravity and metal-poor peculiar classes, for which the current training distribution lacks continuity across some spectral types.[7,8,14,22,23,40] Additional observed spectra would also help reduce the dependence on synthetic augmentation derived from individual spectral standards. A controlled SNR experiment, similar to that of Zhou et al.,[14] should be performed using the present feature representation to quantify the effects of noise on both spectral-type and peculiar-class classification. Repeated stratified partitions or nested model-selection procedures could also be used to evaluate the robustness of the selected configurations while preserving an independent observed test set for final performance assessment. Although the framework was developed using SpeX prism spectra, it could be adapted to other low-resolution near-infrared datasets, provided that the classifiers are retrained using representative standards from the target instrument.

5.1 Acknowledgments

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Data and Code Availability

The near-infrared spectra analyzed in this study are publicly available through the SpeX Prism Library and the SpeX Prism Library Analysis Toolkit (SPLAT) at `https://github.com/aburgasser/splat`.[19] The preprocessed data sets and machine-learning resources developed by Zhou et al. are publicly available through the SpecAI repository at `https://github.com/StellarDataLab/SpecAI`.[14]

The data products generated in the present study are publicly available at `https://github.com/AnaOjedaVera/SpeX-Peculiar-Type-Classification`. The repository also contains the analysis code used to train and evaluate the RF, SVM, KNN, and CNN classifiers and instructions for reproducing the reported results.

The spectra were obtained with the SpeX spectrograph at the NASA Infrared Telescope Facility (IRTF).[18] The analysis was conducted using pandas,[44] NumPy,[45] Matplotlib,[46] and scikit-learn.[30] The CNN was implemented in TensorFlow[31] using the Keras API.[32]

ORCID iDs

Ana V. Ojeda https://orcid.org/0009-0002-0748-9630
Christopher A. Theissen https://orcid.org/0000-0002-9807-5435
C. A. Guerrero https://orcid.org/0000-0002-1113-7489
Juan J. Tapia https://orcid.org/0000-0002-4214-7540
T. Zhou https://orcid.org/0000-0002-3211-2213

APPENDIX A. MANUAL SLOPE SELECTION FOR M- AND L-TYPE STANDARDS

Figure 16 presents the manually selected slope intervals over the mean normalized spectra of the remaining spectral-standard sequences. Panel (a) shows the M0–M9 standards, while panel (b) shows the L0–L9 standards. The same principal and local wavelength intervals described in Sec. 2.3.1 were applied to both sequences.

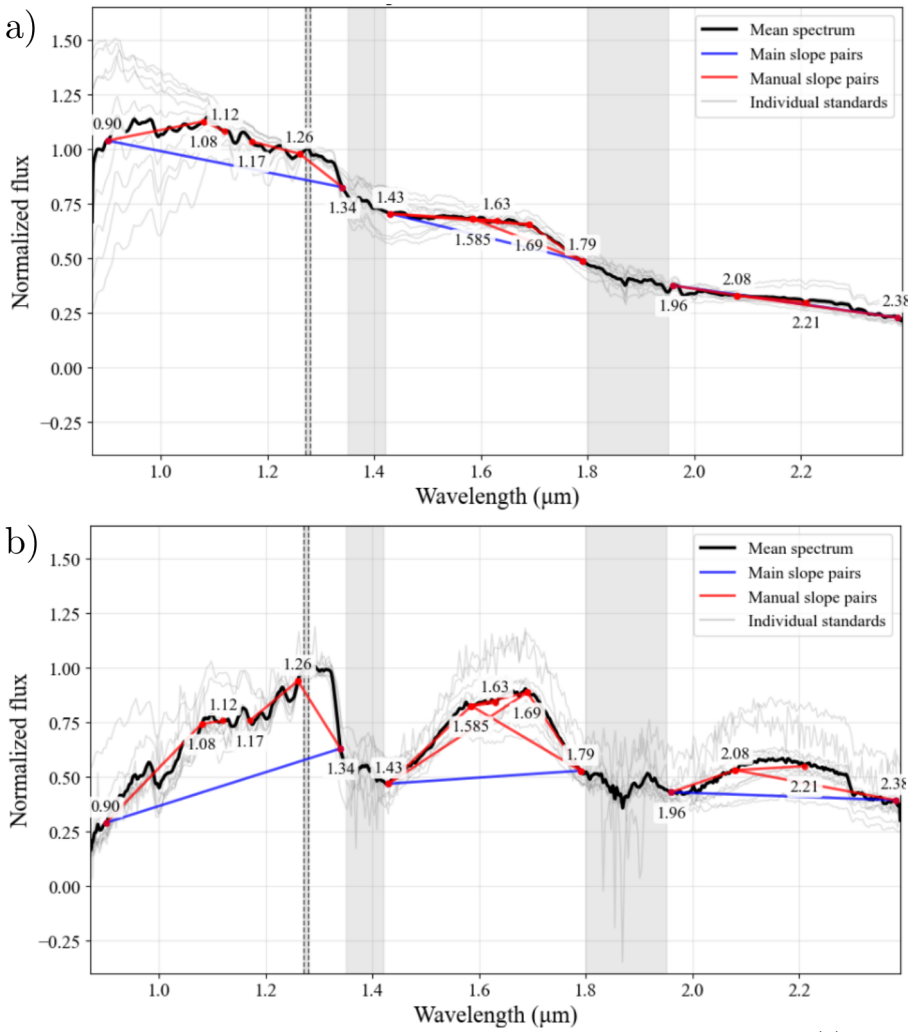


Figure 16: Manually selected spectral slope intervals for the mean normalized spectra of the (a) M0–M9 and (b) L0–L9 spectral standards. The shaded regions indicate the masked telluric absorption intervals.

APPENDIX B. SPECTRAL-TYPE CONFUSION MATRICES

The row-normalized spectral-type confusion matrices for the selected RF, SVM, KNN, and CNN classifiers are presented in Figs. 17–20.

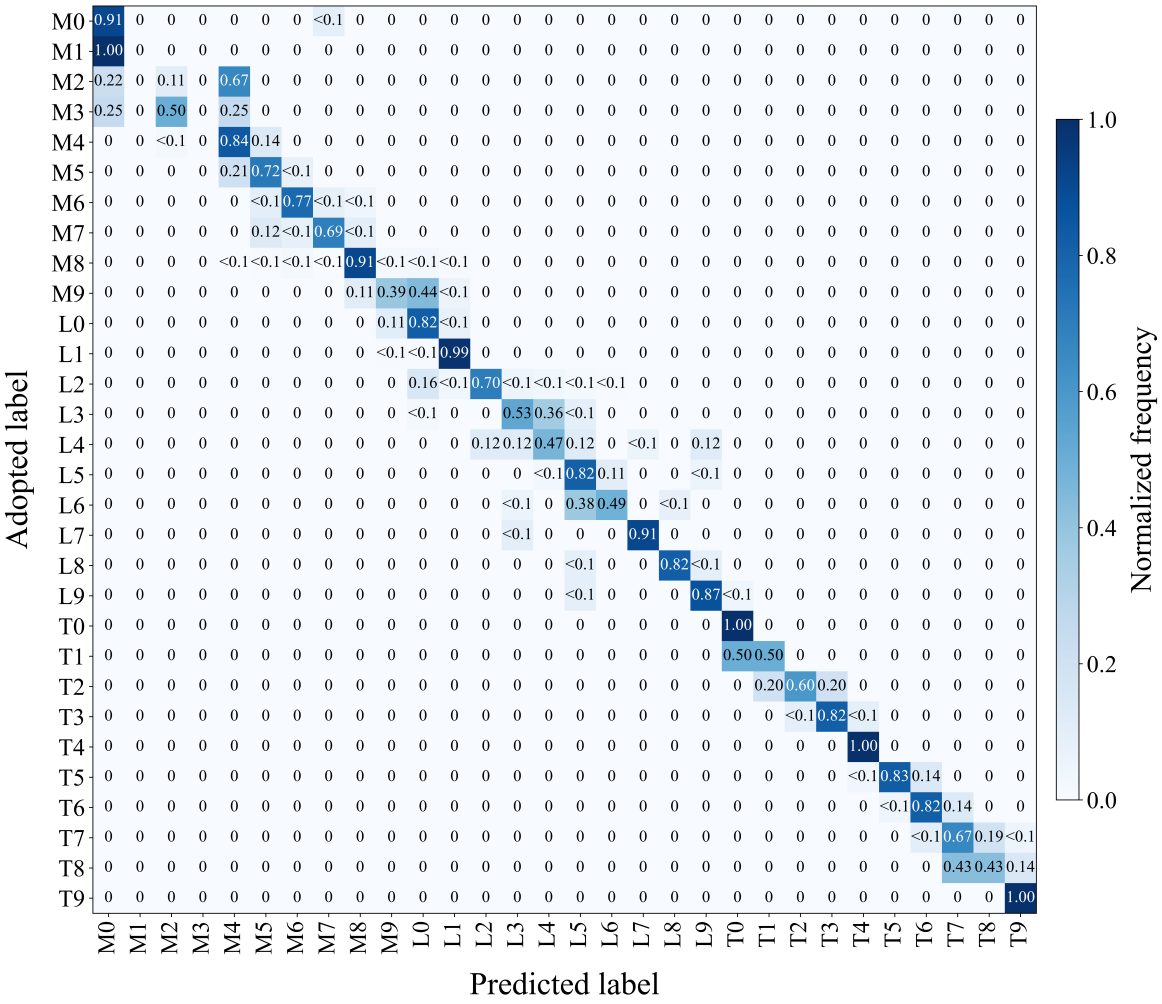


Figure 17: Row-normalized spectral-type confusion matrix for the selected RF classifier on the independent test set. Rows correspond to the adopted M0–T9 spectral types, and columns correspond to the predicted spectral types. Diagonal entries represent exact spectral-type recall, whereas off-diagonal entries show the normalized frequencies of spectral-type misclassifications.

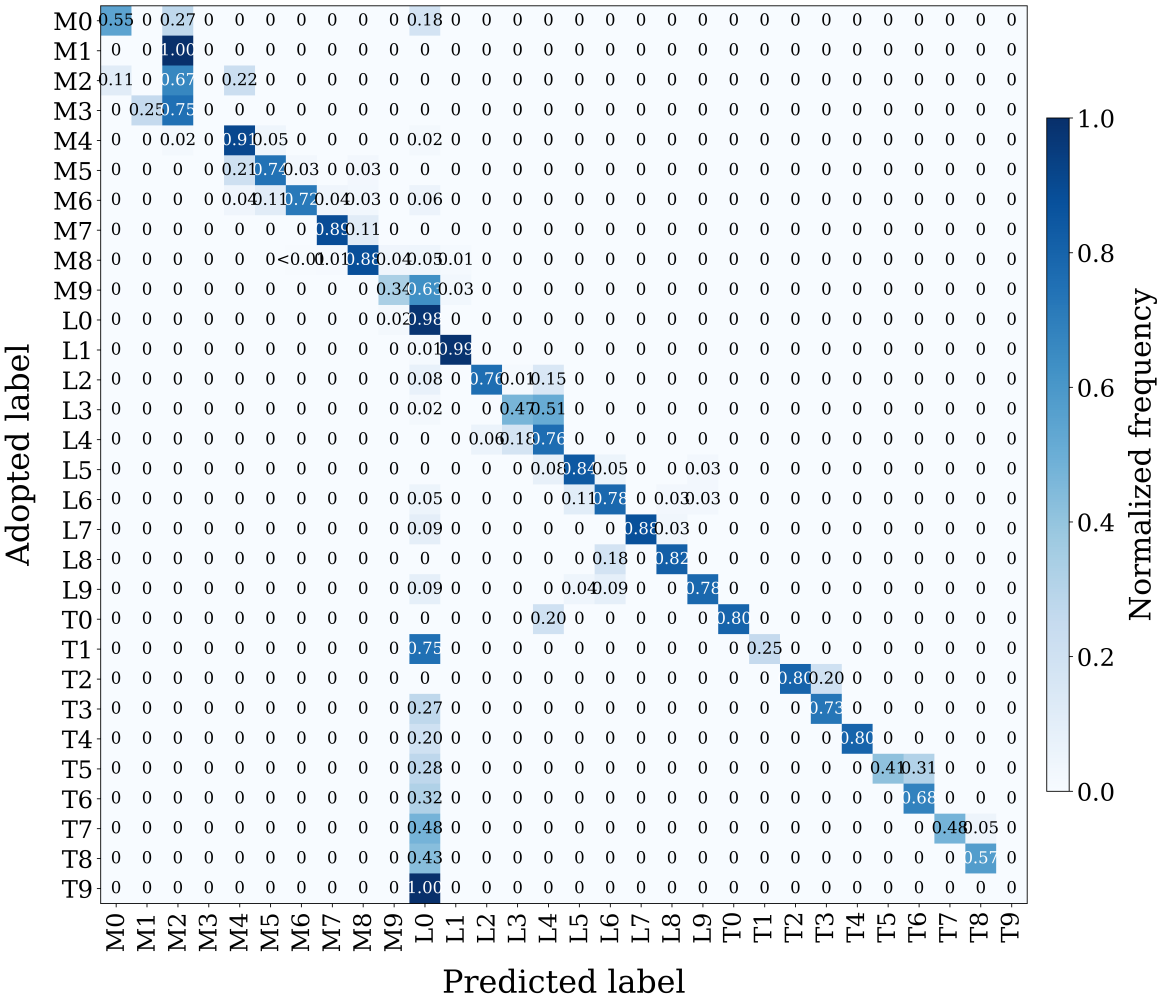


Figure 18: Same as Fig. 17, but for the selected SVM classifier.

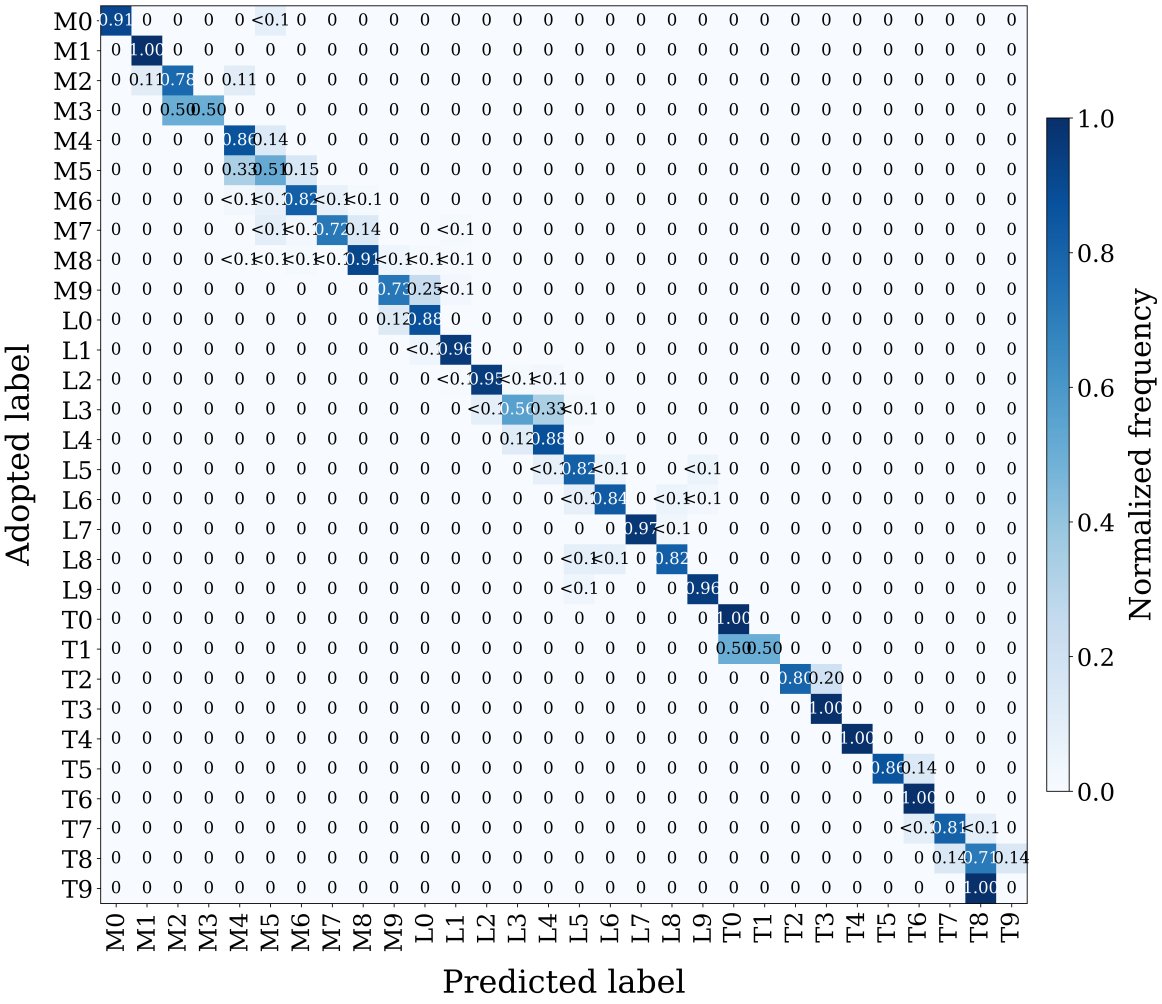


Figure 19: Same as Fig. 17, but for the selected KNN classifier.

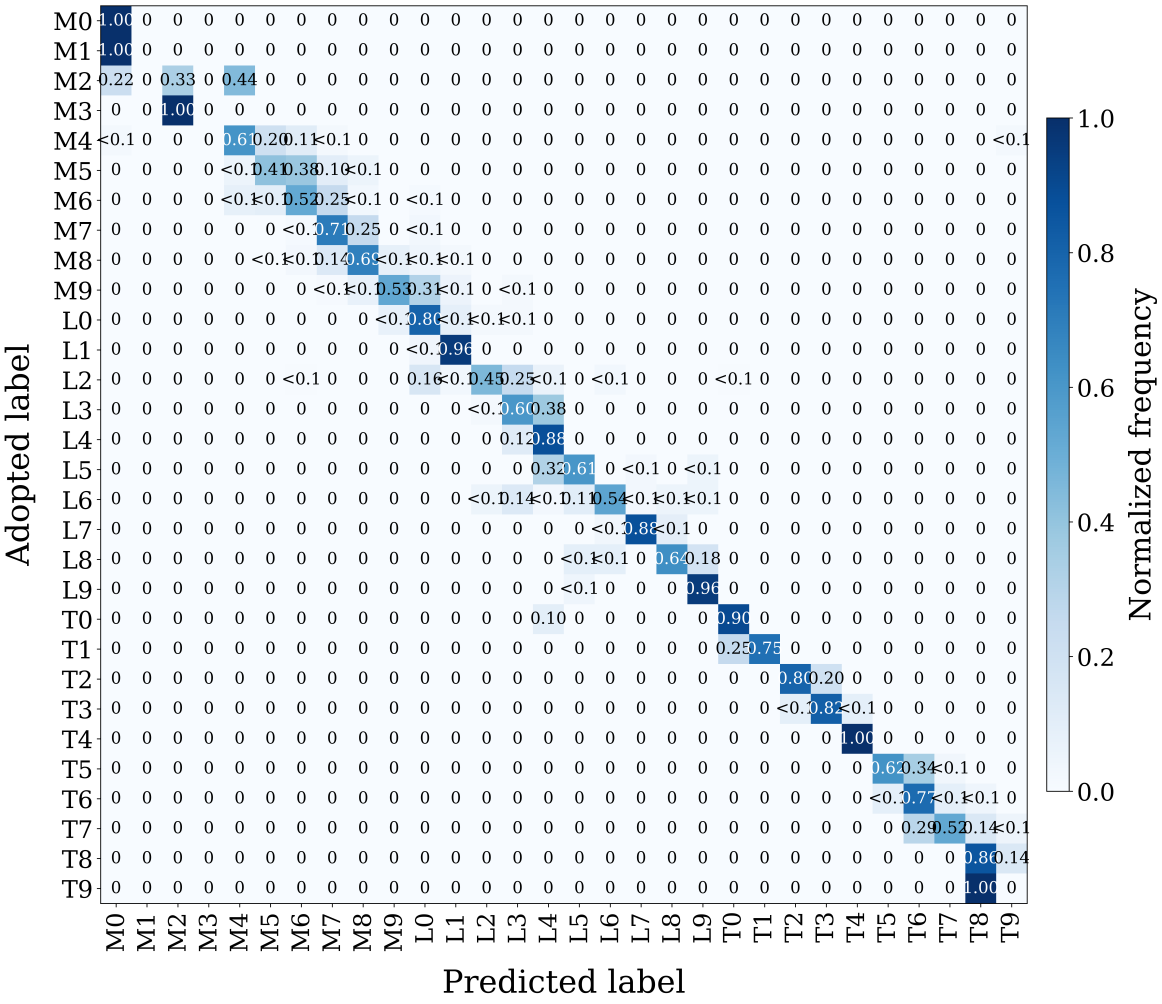


Figure 20: Same as Fig. 17, but for the selected CNN classifier.

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